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A Warp Drive with Predominantly Positive Invariant Energy Density and Global Hawking–Ellis Type I

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Abstract

We present the first fully explicit, continuous, analytically derived warp–drive spacetime within General Relativity whose shift–vector flow is kinematically irrotational. Building on Santiago *et al.* that scalar–potential, zero–vorticity warp fields are Hawking–Ellis Type I for unit lapse and flat spatial slices, we supply a closed–form scalar potential and smooth shift components with proper boundary behavior, together with a Cartan–tetrad analytic pipeline and high–precision eigenanalysis. Compared with the Alcubierre and Natário models (evaluated at identical $(\boldsymbol{\rho}, \boldsymbol{\sigma}, \boldsymbol{v/c})$), our irrotational solution exhibits *significantly reduced* local NEC/WEC stress: its peak proper–energy deficit is reduced by a factor of $\approx \mathbf{38}$ relative to Alcubierre and $\approx \mathbf{2.6} \times \mathbf{10^3}$ relative to Natário, and its peak NEC violation is more than $\mathbf{60}\times$ smaller than Natário. Crucially, the stress–energy is *globally* Hawking–Ellis Type I, with a well–defined timelike eigenvalue (proper energy density) everywhere. A fixed–smoothing vortical ablation confirms that this improvement is causally due to irrotational, curl–free kinematics rather than profile shaping: adding modest vorticity collapses the $\boldsymbol{E_+/E_-}$ balance and drives large increases in the negative–energy magnitude $\boldsymbol{E_-}$. We quantify the negative–energy requirement via a *slice-integrated* (on $\boldsymbol{\Sigma_t}$) negative–energy volume and tabulate global measures. A far–field extrapolation to $\boldsymbol{R \to \infty}$ yields tail–corrected totals $|\boldsymbol{E_+ - E_-}|/(\boldsymbol{E_+ + E_-}) = \mathbf{0.04\%}$. Thus the net *proper* energy is consistent with zero to four decimal places (in fractional units). We also establish regularity at $\boldsymbol{r = 0}$ for the irrotational construction.

Keywords: Warp drive, Energy condition, Hawking–Ellis classification, General Relativity

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1 Introduction

The theoretical investigation of warp–drive spacetimes in General Relativity began with Alcubierre's proposal [1], which produces apparent faster-than-light travel by contracting space ahead of a "bubble" and expanding it behind. While mathematically consistent, the construction requires exotic matter violating the weak and null energy conditions (WEC/NEC), challenging physical plausibility.

Natário [2] introduced zero-expansion warp drives, characterized (under unit lapse and time-independent slicing) by vanishing spatial divergence of the shift, $K = 0 \Leftrightarrow D_i\beta^i = 0$. These models still exhibit NEC violations.

Kinematically irrotational drives (vanishing spatial vorticity, $\boldsymbol{\omega} = 0$) were analyzed by Santiago et al. [3], who showed that for unit lapse ($\alpha = 1$) and flat, time-independent slices the momentum density vanishes and the stress–energy is Hawking–Ellis Type I; they also derived bounds on local energy-condition deficits. Schuster et al. [4] complemented this with global ADM-mass analyses, identifying generic pathologies for typical bubbles. Other approaches include discontinuous or distributional set-ups (e.g., Lentz's phase-engineered profiles [5] and thin-shell models [6]), which entail junction terms; a "tilted" warp paradigm with nonzero vorticity and acceleration [7]; and broader explorations of backgrounds, numerics, and matter models [8–14].

Despite these advances, the literature has lacked a *fully explicit*, continuous, analytically derived construction of a kinematically irrotational warp drive with a closed-form scalar potential and continuous shift components. *Critically, this gap in specific solutions is compounded by a methodological limitation:* many prior studies emphasize observer- or slice-dependent energy diagnostics rather than invariants derived from the stress–energy eigenstructure.

This work. We present, to our knowledge, the first explicit scalar potential $\Phi(r, \theta, t)$ that realizes a continuous, globally regular, kinematically irrotational warp-drive flow with correct asymptotics, satisfying $\boldsymbol{\omega} = *\,\mathrm{d}\boldsymbol{\beta}^\flat = 0$. Irrotationality enforces $\boldsymbol{\beta}^\flat = -\mathrm{d}\Phi$ on each slice and, in the flat static triads adopted here, implies $G_{\hat{0}\hat{i}} = 0$; consequently $G^{\hat{\mu}}{}_{\hat{\nu}}$ is block-diagonal with a unique timelike eigenvalue (the proper energy density), cf. Prop. 2. This algebraic structure accounts for the observed reduction in local NEC/WEC deficits relative to vortical flows. Our results are consistent with and extend prior analyses—recovering the Santiago et al. Type-I mechanism and complementing the ADM-mass considerations of Schuster et al.—while adding an *explicit* construction together with slice-integrated global energy measures (Table 4).

Roadmap and contributions.

We first specify the coordinate system (Fig. 1) and baseline notation; full units and extended conventions are in Appendix A. We then set up the invariant eigenstructure framework for the Einstein tensor—used to extract the proper energy density and Hawking–Ellis type—and summarize the high-precision Cartan–tetrad pipeline employed throughout. Next, we present the explicit scalar-potential construction of the kinematically irrotational warp drive and its kinematic rationale ($\mathrm{d}\boldsymbol{\beta}^\flat = 0 \Rightarrow G_{\hat{0}\hat{i}} = 0$), followed by algebraic and energy-condition maps, whole-slice global budgets, and head-to-head comparisons with the Alcubierre and Natário models. The main contributions

are: (i) an explicit, continuous analytic construction of a kinematically irrotational warp drive (closed-form Φ and continuous β^i); (ii) a high-precision Cartan–tetrad pipeline that extracts invariant proper energy density via the timelike eigenvalue and cross-checks it against a Hessian invariant; (iii) full-spacetime Hawking–Ellis classification maps and quantitative energy-condition diagnostics, showing substantially reduced local NEC/WEC deficits relative to vortical flows; (iv) global, slice-integrated energy budgets with tail modeling, indicating near-cancellation of proper energy (baseline window and tail-extrapolated totals); (v) a consolidated algebraic type picture under the stated assumptions ($\alpha = 1$, flat, time-independent slices): the irrotational geometry is globally Type I in this setup; Alcubierre and Natário exhibit a mix of Types I and IV, with no Type II/III regions observed.

2 Timelike eigenvalue definition of proper energy density and rest frame

Quick map. For a compact signature/tensor-form crosswalk see Table 1; for a side-by-side comparison of algebraic features relevant here (diagonalizability, eigenvalue content, existence of a real rest frame, and characteristic energy-condition behavior), see Table 2.

Convention and use. We adopt signature $(-,+,+,+)$ and define the proper energy density via the timelike eigenvalue of the mixed stress–energy tensor $T^{\mu}{}_{\nu}$ in its principal orthonormal frame [15–19]:

$$T^{(\mu)}{}_{(\nu)} \;=\; \mathrm{diag}\big(-\,\varrho_p,\; p_1,\; p_2,\; p_3\big). \tag{1}$$

For comparison with prior literature one may also state the standard eigenvalue problem in the Lichnerowicz [15] convention (often written for $(+,-,-,-)$):

$$T^{\mu}{}_{\nu}\, u^{\nu} \;=\; \varrho_p\, u^{\mu}, \tag{2}$$

while in our adopted $(-,+,+,+)$ signature this reads

$$T^{\mu}{}_{\nu}\, u^{\nu} \;=\; -\,\varrho_p\, u^{\mu}. \tag{3}$$

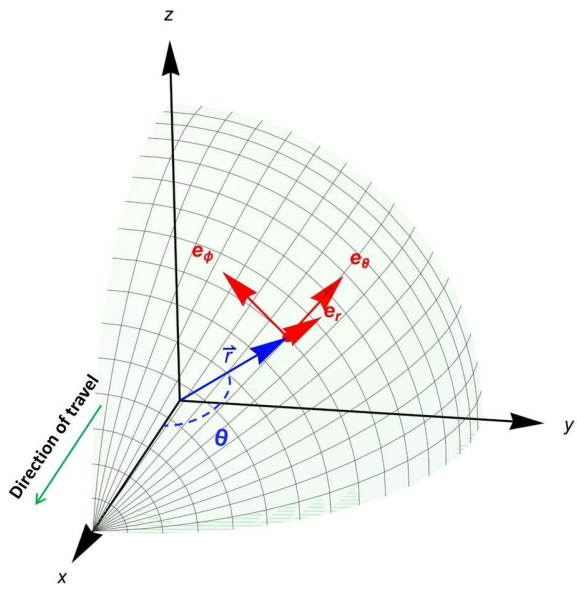
Equivalently, the covariant non-standard form becomes the generalized eigenproblem

$$T_{\mu\nu}\, u^{\nu} \;=\; -\,\varrho_p\, g_{\mu\nu}\, u^{\nu}. \tag{4}$$

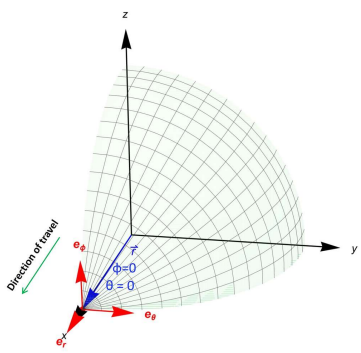
When a real timelike eigenvector exists the tensor is Hawking–Ellis Type I and admits a unique rest frame; in Type IV a complex conjugate pair appears and no real rest frame exists. In the principal frame the null and weak energy conditions take the standard forms (for all i):

$$\text{NEC}: \qquad \varrho_p + p_i \;\geq\; 0, \tag{5}$$

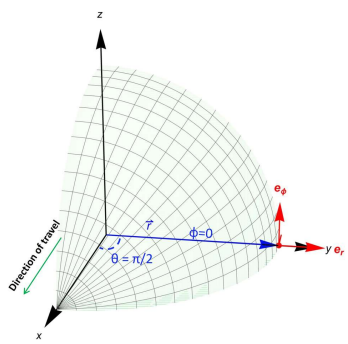
$$\text{WEC}: \qquad \varrho_p \;\geq\; 0 \quad \text{and} \quad \varrho_p + p_i \;\geq\; 0. \tag{6}$$



(a) P at $(r = \rho,\ 0 < \theta < \pi/2,\ 0 < \phi < \pi/2)$, with θ measured from $+x$ and ϕ in the yz (equatorial) plane from $+y$ toward $+z$



(b) P at $(r = \rho,\ \theta = 0,\ \phi = 0)$



(c) P at $(r = \rho,\ \theta = \pi/2,\ \phi = 0)$

Fig. 1: Natário's spherical coordinate system (r, θ, ϕ) for the warp drive moving in the x direction.

Tensor Form	Signature	Proper Energy Density (ϱ_p)	Principal Pressures (p_i)
Mixed ($T^{(\mu)}{}_{(\nu)}$)	$(-,+,+,+)$	$-T^{(0)}{}_{(0)}$	$T^{(i)}{}_{(i)}$
Mixed ($T^{(\mu)}{}_{(\nu)}$)	$(+,-,-,-)$	$T^{(0)}{}_{(0)}$	$-T^{(i)}{}_{(i)}$
Covariant ($T_{(\hat{\mu})(\hat{\nu})}$)	$(-,+,+,+)$	$T_{(\hat{0})(\hat{0})}$	$T_{(\hat{i})(\hat{i})}$
Covariant ($T_{(\hat{\mu})(\hat{\nu})}$)	$(+,-,-,-)$	$T_{(\hat{0})(\hat{0})}$	$T_{(\hat{i})(\hat{i})}$
Contravariant ($T^{(\hat{\mu})(\hat{\nu})}$)	$(-,+,+,+)$	$T^{(\hat{0})(\hat{0})}$	$T^{(\hat{i})(\hat{i})}$
Contravariant ($T^{(\hat{\mu})(\hat{\nu})}$)	$(+,-,-,-)$	$T^{(\hat{0})(\hat{0})}$	$T^{(\hat{i})(\hat{i})}$

Table 1: Proper energy density ϱ_p and principal pressures p_i in the principal orthonormal frame, showing their relation to diagonal tensor components for various tensor forms and metric signatures (*assuming real eigenvalues exist*).

Table 2: Algebraic Comparison of Hawking–Ellis Type I and Type IV Stress-Energy Tensors.

Property	**Type I** (e.g., Classical Fluids and Non-null Fields)	**Type IV** (e.g., Alcubierre & Natário-Zero-Exp. Warp Drives)
Eigenvalues of $T^{\mu}{}_{\nu}$	ONE real timelike ($-\varrho_p$), THREE real spacelike (p_i)	ONE pair complex conjugate, TWO real spacelike
Diagonaliz. & Rest Frame	Diagonalizable over $\mathbb{R}$. Admits a unique, real orthonormal tetrad frame (rest frame) where $T^{\hat{\mu}}{}_{\hat{\nu}} = \mathrm{diag}(-\varrho_p, p_1, p_2, p_3)$.	Diagonalizable only over $\mathbb{C}$. Does not admit a real orthonormal tetrad rest frame; eigenvectors associated with complex eigenvalues are complex conjugates.
Physical Nature & Energy Conditions	Typically represents classical matter/energy distributions. Can satisfy standard energy conditions (e.g., NEC, WEC).	Represents non-classical, exotic distributions. Characteristically violates all standard energy conditions (e.g., NEC, WEC) [20, 21]. Requires non-static spacetime in $f(R)$ gravity [22].

3 Warp-drive spacetimes in a spherical coordinate tetrad framework

In this section, we systematically analyze warp-drive spacetimes within a spherical coordinate orthonormal tetrad framework, providing a clear and physically insightful decomposition of their kinematic and dynamic properties. We begin by discussing the generalized warp-drive spacetime, establishing a unified mathematical setting suitable for encompassing various specific warp-drive geometries and facilitating comparisons among them. Subsequently, we focus on the well-known Alcubierre warp drive, explicitly recasting its geometry and stress-energy content into spherical orthonormal tetrad components, thereby clarifying its structure and physical interpretation. We then explore Natário's kinematically divergenceless (zero expansion) warp drive, highlighting its distinct geometric features. Lastly, we address kinematically irrotational warp drives, examining the rationale for considering vorticity-free conditions in warp-drive

models, and analyzing their physical consequences within our tetrad formalism. For a compact side-by-side comparison of the three families we study, including defining kinematic constraints and shift components, see Table 3.

3.1 The generalized warp drive

In this subsection, we introduce Natário's generalized warp-drive spacetime within the 3+1 ADM decomposition framework. We establish the mathematical foundations necessary to describe arbitrary warp-drive configurations, emphasizing the role of the lapse and shift functions in characterizing the geometry and evolution of spacetime. We then define the line element of the generalized warp drive, clearly delineating the distinction between coordinate and tetrad-based descriptions. This sets the stage for comparing specialized warp-drive models, such as Alcubierre's original spacetime, Natário's zero-expansion configuration, and a new irrotational warp drive which will be discussed in subsequent subsections.

Natário [2] characterized a generalized warp-drive spacetime, whose geometry is associated with a vector field $\mathbf{X}$, using the 3+1 Arnowitt-Deser-Misner (ADM) formalism (Appendix A) [23–26]. In this framework, spacetime is foliated by a sequence of spacelike hypersurfaces Σ_t. The *Eulerian observers* are defined by the condition that their 4-velocity n^α is the future-pointing unit vector field normal to these hypersurfaces. The vector field $\mathbf{X}$ generating the warp effect is identified with the ADM shift vector $\boldsymbol{\beta}$, as discussed below (7). A key consideration is the causal structure: while such spacetimes may satisfy global hyperbolicity for subliminal speeds, superluminal warp drives generally exhibit Cauchy horizons [27], restricting the *global* applicability of the 3+1 decomposition, although it typically remains valid *locally*.

We use the standard symbols α (lapse) and $\boldsymbol{\beta}$ (shift), but adopt the *minus-sign* shift convention in the line element, i.e. $(dx^i - \beta^i dt)$ rather than $(dx^i + \beta^i dt)$. Relative to Gourgoulhon [24] and Alcubierre [25], this corresponds to $\beta_i^{(\text{here})} = -\beta_i^{(\text{there})}$. For $\alpha = 1$ the slice normal has $n_\mu = (-1, 0, 0, 0)$ and $n^\mu = g^{\mu\nu} n_\nu = (1, \beta^i)$ (since $g^{0i} = -\beta^i$ in this convention), and the time vector decomposes as $\partial_t = \alpha n - \beta^i \partial_i = n - \beta^i \partial_i$. (Misner et al. [23] use N and N^i for lapse and shift.)

Following Natário [2] and common practice for the warp-drive models considered here, we set the lapse function $\alpha = 1$, simplifying the relation between coordinate time t and proper time τ along the normal direction ($d\tau = \alpha\, dt = dt$).

The shift vector $\boldsymbol{\beta}$ measures how spatial coordinates are displaced along $\boldsymbol{\beta}$ between adjacent hypersurfaces Σ_t relative to the normal vector [24]. The sign of its components determines the orientation (e.g., positive or negative x-direction) of this coordinate shift, representing a local velocity field relative to Eulerian observers at rest in the slicing. Following convention [3], we relate the generating vector field $\mathbf{X}$ directly to the shift vector as

$$\boldsymbol{\beta} = \mathbf{X}. \tag{7}$$

This choice corresponds to a *flow* interpretation. The common negative sign convention $\beta^i = X^i = -v^i$ often seen in the literature [3] typically relates the shift vector β^i (local coordinate flow velocity relative to Eulerian observers) to the bubble's coordinate velocity v^i as measured by distant observers, with the sign arising from the model's specific dynamics.

With these choices ($\alpha = 1$, $\gamma_{ij} = \delta_{ij}$ for flat spatial slices in Cartesian coordinates), the line element with signature $(-,+,+,+)$ for a *generalized warp-drive spacetime* takes the form [2]:

$$\begin{aligned} ds^2 &= -dt^2 + \delta_{ij}\big(dx^i - \beta^i dt\big)\big(dx^j - \beta^j dt\big) \\ &= -dt^2 + \big(dx - \beta^x dt\big)^2 + \big(dy - \beta^y dt\big)^2 + \big(dz - \beta^z dt\big)^2. \end{aligned} \tag{8}$$

where $(x^1, x^2, x^3) = (x, y, z)$ are orthogonal Cartesian coordinates, and $\beta^i = (\beta^x, \beta^y, \beta^z)$ are the corresponding Cartesian coordinate components of the shift vector field $\boldsymbol{\beta}$.

To facilitate physical interpretation and analysis, particularly in scenarios with spherical symmetry or specific directional properties, it is advantageous to work with local orthonormal frames (tetrads or vierbein), wherein the metric components are locally Minkowski ($\eta_{\hat{\alpha}\hat{\beta}}$) [17, 23, 28–30]. In spherical coordinates (r, θ, ϕ) (defined in Appendix A), the spatial unit orthonormal basis vectors $\boldsymbol{e}_{\hat{i}}$ are related to the coordinate basis vectors $\boldsymbol{g}_i = \partial/\partial x^i$ by:

$$\boldsymbol{e}_{\hat{r}} \equiv \frac{\partial}{\partial r} = \boldsymbol{g}_r \ , \ \boldsymbol{e}_{\hat{\theta}} \equiv \frac{1}{r}\frac{\partial}{\partial \theta} = \frac{1}{r}\boldsymbol{g}_\theta \ , \ \boldsymbol{e}_{\hat{\phi}} \equiv \frac{1}{r\sin\theta}\frac{\partial}{\partial \phi} = \frac{1}{r\sin\theta}\boldsymbol{g}_\phi \, . \tag{9}$$

Relative to this basis, the shift vector $\boldsymbol{\beta}$ is then expressed as:

$$\boldsymbol{\beta} = \beta^{\hat{r}}\,\boldsymbol{e}_{\hat{r}} + \beta^{\hat{\theta}}\,\boldsymbol{e}_{\hat{\theta}} + \beta^{\hat{\phi}}\,\boldsymbol{e}_{\hat{\phi}} \, , \tag{10}$$

where $\beta^{\hat{i}}$ are the tetrad (physical) components.

We adopt the convention of using carets (hats) for tetrad indices [23]. The corresponding dual basis one-forms are $\boldsymbol{e}^{\hat{r}} = \mathrm{d}r$, $\boldsymbol{e}^{\hat{\theta}} = r\,\mathrm{d}\theta$, $\boldsymbol{e}^{\hat{\phi}} = r\sin\theta\,\mathrm{d}\phi$. While the radial basis one-form $\boldsymbol{e}^{\hat{r}}$ is exact, the angular one-forms are not closed ($\mathrm{d}\boldsymbol{e}^{\hat{\theta}} \neq 0, \mathrm{d}\boldsymbol{e}^{\hat{\phi}} \neq 0$), reflecting the fact that this standard spherical orthonormal basis is non-holonomic, meaning it cannot be integrated to define a coordinate chart on the differentiable manifold. Equivalently, the basis vector fields do not commute, as shown by the Lie bracket: e.g., $[\boldsymbol{e}_{\hat{\theta}}, \boldsymbol{e}_{\hat{\phi}}] = -\frac{\cot\theta}{r}\,\boldsymbol{e}_{\hat{\phi}} \neq 0$ (transporting a vector around an infinitesimal loop in this plane results in a net rotation of the vector).

The components of the shift vector $\boldsymbol{\beta}$ in the coordinate bases $\boldsymbol{g}_i$ and orthonormal bases $\boldsymbol{e}_{\hat{i}}$ are related via the definitions in (9). For instance, $\beta^{\hat{r}} = \beta^r$, $\beta^{\hat{\theta}} = r\beta^\theta$, $\beta^{\hat{\phi}} = r\sin\theta\beta^\phi$. Since the spatial part of the tetrad metric is $\delta_{\hat{i}\hat{j}}$, the distinction between covariant and contravariant spatial tetrad components is trivial ($\beta_{\hat{i}} = \delta_{\hat{i}\hat{j}}\beta^{\hat{j}} = \beta^{\hat{i}}$).

Tetrad orthonormal basis components represent the measurements an observer in free fall (subject only to gravity) would record—that is, the *physical components* in a local Minkowski frame. No single inertial frame can globally encompass the effects of spacetime curvature; however, in a sufficiently small region (a local Lorentz frame), the equivalence principle ensures that Special Relativity applies.

Moreover, although tetrad components are the components of tensors relative to a local orthonormal basis, they do not transform under general coordinate transformations in the same way as coordinate basis components. Instead, when the tetrad basis is changed, the components transform via local Lorentz transformations.

Finally, the tetrad formalism employs the Ricci rotation coefficients (or spin connection), which encode how the tetrad fields rotate relative to the Levi–Civita connection as one moves through spacetime. These coefficients reflect the geometric obstruction to defining a global inertial frame in a curved spacetime.

3.2 Alcubierre's warp drive in a spherical coordinate orthonormal tetrad

Alcubierre's solution [1] was originally written in Cartesian (t, x, y, z) coordinates. Because the geometry is axially symmetric about the $+x$ axis (direction of travel), it is convenient to adopt the spherical chart (t, r, θ, ϕ) shown in Fig. 1 and to pass to an orthonormal tetrad $\{\boldsymbol{e}_{\hat{0}}, \boldsymbol{e}_{\hat{r}}, \boldsymbol{e}_{\hat{\theta}}, \boldsymbol{e}_{\hat{\phi}}\}$, with dual one-forms $\{\boldsymbol{e}^{\hat{0}}, \boldsymbol{e}^{\hat{r}}, \boldsymbol{e}^{\hat{\theta}}, \boldsymbol{e}^{\hat{\phi}}\}$. We work in geometric units with $c = 1$ unless stated otherwise. We leave Einstein's coupling constant κ explicit.

Shift vector

Rotational symmetry implies $\beta^{\hat{\phi}} = 0$; hence

$$\boldsymbol{\beta} = \beta^{\hat{r}}\, \boldsymbol{e}_{\hat{r}} + \beta^{\hat{\theta}}\, \boldsymbol{e}_{\hat{\theta}}\,. \tag{11}$$

Following Alcubierre we choose the spatial covector $\boldsymbol{\beta}^\flat = v(t)\, f_{\mathrm{Alc}}(r)\, \mathrm{d}x$ with $x = r\cos\theta$. Using $\mathrm{d}x = \cos\theta\, \mathrm{d}r - r\sin\theta\, \mathrm{d}\theta$ and the spherical dual basis $\boldsymbol{e}^{\hat{r}} = \mathrm{d}r$, $\boldsymbol{e}^{\hat{\theta}} = r\, \mathrm{d}\theta$, one finds

$$\boldsymbol{\beta}^\flat = v(t)\, f_{\mathrm{Alc}}(r) \big(\cos\theta\, \boldsymbol{e}^{\hat{r}} - \sin\theta\, \boldsymbol{e}^{\hat{\theta}}\big). \tag{12}$$

Because the triad $\{\boldsymbol{e}_{\hat{r}}, \boldsymbol{e}_{\hat{\theta}}, \boldsymbol{e}_{\hat{\phi}}\}$ is *spatial* and orthonormal, raising indices changes only the sign of the time component; therefore $\beta^{\hat{i}} = \beta_{\hat{i}}$ for $i = r, \theta, \phi$:

$$\begin{pmatrix} \beta^{\hat{r}} \\ \beta^{\hat{\theta}} \\ \beta^{\hat{\phi}} \end{pmatrix} = v(t)\, f_{\mathrm{Alc}}(r) \begin{pmatrix} \cos\theta \\ -\sin\theta \\ 0 \end{pmatrix}. \tag{13}$$

Form function

The smooth *top-hat* function

$$f_{\mathrm{Alc}}(r) = \frac{\tanh[\sigma(r+\rho)] - \tanh[\sigma(r-\rho)]}{2\, \tanh(\sigma\rho)}, \tag{14}$$

obeys $f_{\mathrm{Alc}}(0) = 1$ and $\lim_{r\to\infty} f_{\mathrm{Alc}}(r) = 0$. Its transition layer of width $\simeq \pm 3/\sigma$ is centered on $r \simeq \rho$. Here ρ is the warp bubble radius and σ^{-1} its wall thickness.

Boundary behaviour (summary)

Inside the bubble the shift approaches $+v(t)$ along $+x$; far away it decays to zero. Detailed small/large–r limits at $\theta = 0$ and $\theta = \pi/2$ are listed in Appendix E.

3.3 Natário's kinematically divergence-free warp drive

Natário [2] sought a warp-drive spacetime with *vanishing spatial expansion*. This coincides with zero kinematic divergence $\operatorname{div}\boldsymbol{\beta} = 0$ under unit lapse, flat time-independent 3+1 slicing. Working on each constant-time slice $(\mathbb{R}^3, \gamma)$ one may write

$$\operatorname{div}\boldsymbol{\beta} = \frac{1}{\sqrt{\gamma}}\partial_i\big(\sqrt{\gamma}\,\beta^i\big) = -\delta\boldsymbol{\beta}^\flat = 0, \tag{15}$$

where δ is the three–dimensional exterior codifferential. Hence $\boldsymbol{\beta}^\flat$ must be *co-exact*:

$$\boldsymbol{\beta}^\flat = *\mathrm{d}\boldsymbol{A} \qquad (\exists\, 1\text{-form } \boldsymbol{A}). \tag{16}$$

Choosing $\boldsymbol{A} = -\frac{1}{2}\,f(r)\,r^2\sin^2\theta\,\mathrm{d}\phi$ and multiplying by an arbitrary bubble velocity $v(t)$ gives

$$\boldsymbol{\beta}^\flat = -v(t)\;*\,\mathrm{d}\big[f(r)\,\tfrac{1}{2}r^2\sin^2\theta\,\mathrm{d}\phi\big]. \tag{17}$$

A short calculation produces covariant components

$$\beta_{\hat{r}} = -v(t)\,f(r)\cos\theta, \tag{18}$$

$$\beta_{\hat{\theta}} = v(t)\Big(f(r) + \tfrac{r}{2}f'(r)\Big)\sin\theta, \tag{19}$$

$$\beta_{\hat{\phi}} = 0, \tag{20}$$

so that in our spatial orthonormal triad

$$\begin{pmatrix}\beta^{\hat{r}}\\ \beta^{\hat{\theta}}\\ \beta^{\hat{\phi}}\end{pmatrix} = \begin{pmatrix} -\,v(t)\,f(r)\cos\theta \\ v(t)\,\big(f(r) + \frac{r}{2}f'(r)\big)\sin\theta \\ 0\end{pmatrix}. \tag{21}$$

Form function for Natário

To facilitate comparison with Alcubierre we define

$$f(r) := 1 - f_{\mathrm{Alc}}(r), \tag{22}$$

so that $f(0) = 0$ and $\lim_{r\to\infty} f(r) = 1$. With this choice the Eulerian observer at the bubble center ($r = 0$) sees distant matter move with speed $-v(t)$, opposite to Alcubierre's convention.

Boundary behaviour (summary)

The center of the bubble is at rest while the distant universe drifts with velocity $-v(t)$, consistent with Natário's convention. Detailed limits appear in Appendix E.

Kinematic constraint and global energy conditions

Natário's divergence–free ansatz ($K \equiv \gamma^{ij}K_{ij} = 0$, equivalently $\mathrm{div}_{\Sigma_t}\boldsymbol{\beta} = 0$ for unit lapse and time-independent spatial metric) renders every time slice Σ_t *instantaneously* isochoric. Yet if the driving function $v(t)$ varies in time the resulting shift field $\beta^i(t, x^k)$ makes the spacetime *non-stationary*. Although $K = 0$ on each slice, the individual components

$$K_{ij} \;=\; \tfrac{1}{2}\big(D_i\beta_j + D_j\beta_i\big) \tag{23}$$

acquire explicit t–dependence, and so do the curvature scalars that enter the mixed Einstein tensor $G^\mu{}_\nu$.

The averaged null energy condition (ANEC) probes the *integrated* null projection,

$$\mathscr{I}_{\mathrm{ANEC}} \;=\; \int_\gamma T_{\mu\nu}k^\mu k^\nu \,\mathrm{d}\lambda = \frac{1}{\kappa}\int_\gamma G_{\mu\nu}k^\mu k^\nu \,\mathrm{d}\lambda, \tag{24}$$

taken along complete null geodesics k^μ. Because such geodesics thread a sequence of slices that may evolve in time, a quantity that vanishes *instantaneously* on each Σ_t—for example the local expansion K—may still yield a non-zero contribution to the integral, with sign depending on the detailed evolution. Thus, imposing $K = 0$ on each slice does not by itself establish compliance with averaged energy conditions. In practice, one should assess the full four-dimensional evolution of $\big(\alpha, \beta^i, \gamma_{ij}\big)$ along the null geodesics when evaluating ANEC (and related *averaged* conditions).

Proposition 1 (Trace–energy condition for the Natário warp drive)**.** ***Assumptions:*** *$\alpha \equiv 1$, flat* and static *spatial slice ($R^{(3)} = 0$), vanishing expansion $K = 0$, and $\mathscr{L}_n K = 0$* ($K = 0$ fixes only the slice value, whereas $\mathscr{L}_n K$ is its normal–time derivative and can be nonzero).

Then the Einstein–tensor trace satisfies $G \leq 0$ everywhere; equivalently, the trace–energy condition $T \leq 0$ holds throughout the spacetime.

Proof. Work in a 3+1 decomposition with $(-,+,+,+)$ signature, $\gamma_{ij} \simeq \mathbb{R}^3$ and $K_{ij} = -\frac{1}{2}\mathscr{L}_n\gamma_{ij}$. The Gauss–Codazzi scalar identity gives

$$^{(4)}R = {}^{(3)}R + K^2 + K_{ij}K^{ij} - \frac{2}{\alpha}\,\mathscr{L}_n K - \frac{2}{\alpha}\,D_i D^i\alpha. \tag{25}$$

With $R^{(3)} = 0$, $K = 0$, $D_i\alpha = 0$ and $\mathscr{L}_n K = 0$, we obtain $R^{(4)} = K_{ij}K^{ij} \geq 0$. Since $G = -R^{(4)}$, one has $G \leq 0$, and thus $T = G/\kappa \leq 0$. □

3.4 Kinematically irrotational warp drives

We now construct a warp-drive spacetime whose *shift* is irrotational in the (3+1) sense, i.e. whose spatial vorticity tensor $\omega_{ij} := \frac{1}{2}\big(D_j\beta_i - D_i\beta_j\big)$ vanishes on every constant-time slice. The discussion is organized in two parts: first a justification for adopting a *kinematic* (purely geometric) definition of vorticity, in preference to fluid-dynamical potential-flow ideas; second the actual derivation of an irrotational shift vector.

3.4.1 Justification for a kinematic–vorticity approach

In warp geometries the "flow" is the geometric shift $\boldsymbol{\beta}$, not a material four–velocity, so fluid–enthalpy analogies do not apply. Under our slice assumptions (unit lapse $\alpha \equiv 1$, flat, time-independent spatial slices), imposing $\mathrm{d}\boldsymbol{\beta}^\flat = 0$ (i.e., $\boldsymbol{\beta}^\flat = -\mathrm{d}\Phi$) sets $G_{\hat{0}\hat{i}} = 0$ (see Eq. (47)) and block-diagonalizes $G^{\hat{\mu}}{}_{\hat{\nu}}$ in the Eulerian tetrad, yielding a unique timelike eigenvector and Hawking–Ellis Type I everywhere for our construction (Prop. 2). This algebraic mechanism—eliminating momentum density and the associated Type IV patches—is the reason to enforce *kinematic* irrotationality.

A fixed–smoothing ablation that adds only a vortical component (with $g(r)$, masks, and tolerances held fixed) shows the causal role: the global budgets degrade sharply (e.g., E_- grows by $> 500\times$ at $\eta = 1$), proving the improved NEC/WEC profile is due to curl-free kinematics rather than profile smoothing (Appendix D, Table D6).

We deliberately *do not* also impose Natário's $\operatorname{div}\boldsymbol{\beta} = 0$ (Sec. 3.3). Enforcing both $\mathrm{d}\boldsymbol{\beta}^\flat = 0$ and $\operatorname{div}\boldsymbol{\beta} = 0$ would force a harmonic potential ($\nabla^2\Phi = 0$), severely restricting localized bubbles; allowing $\operatorname{div}\boldsymbol{\beta} \neq 0$ enlarges the design space and, in our numerics, reduces negative-energy magnitudes.

In summary, within these slice assumptions, irrotationality is a *sufficient* and conceptually clean algebraic constraint for achieving global Type I; we do not claim it is necessary in general.

3.4.2 Kinematically irrotational warp drive

In this section, we construct an irrotational warp-drive spacetime by defining a velocity field using an orthonormal tetrad framework in spherical coordinates. The tetrad components are chosen to exploit the continuous rotational symmetry around the x direction of travel, following a similar approach to that used by Natário in his zero-expansion warp-drive model [2].

In General Relativity, the vorticity characterizes the local rotation of the congruence of timelike worldlines (paths of observers) in spacetime. On a 4-D Lorentzian pseudo-Riemannian manifold (M^4, g), the spacetime vorticity tensor $\omega^{(4)}_{\mu\nu}$ characterizes dynamic rotation associated with a 4-velocity field u^μ. In the 3+1 decomposition, we focus on the kinematic Euclidean $(\mathbb{R}^3,\gamma)$ spatial vorticity tensor on each slice, defined using the spatial covariant derivative D_i acting on the shift vector $\boldsymbol{\beta}$:

$$\omega_{ij} = D_{[i}\beta_{j]} = \frac{1}{2}\left(D_i\beta_j - D_j\beta_i\right). \tag{26}$$

The *physical spacetime vorticity* of a timelike congruence with 4–velocity u^μ is the spatially projected antisymmetric part $\omega^{(4)}_{\mu\nu} = h_\mu{}^\alpha h_\nu{}^\beta \nabla_{[\alpha}u_{\beta]}$, $h_{\mu\nu} = g_{\mu\nu} + u_\mu u_\nu$. In exterior–calculus form this reads $\omega^{(4)} = (1/2)(\mathrm{d}_4 u^\flat - u^\flat \wedge a^\flat)$, $a^\flat := (u\cdot\nabla)u_\mu\,\mathrm{d}x^\mu$,.

By contrast, the *kinematic (purely spatial) vorticity* of the shift on each slice Σ_t is

$$\omega = \mathrm{d}\boldsymbol{\beta}^\flat, \tag{27}$$

where d is the 3D exterior derivative. Our irrotationality condition is $\omega = 0$, which (on simply connected slices) implies $\boldsymbol{\beta}^\flat = -\mathrm{d}\Phi$. This constraint involves only spatial

derivatives; therefore $\omega = 0$ does *not* imply $\omega^{(4)} = 0$ if $\boldsymbol{\beta}$ depends on time, because the acceleration/tilt term $u^\flat \wedge a^\flat$ can remain nonzero.

In the $3+1$ decomposition formalism with lapse function $\alpha = 1$, the coordinate time t measures proper time along the worldlines of observers moving orthogonally to the spatial hypersurfaces (normal observers). Consequently, the lapse function does not contribute to the vorticity, and the dynamics of the flow are entirely encoded in the behavior of the shift vector $\boldsymbol{\beta}$.

Computing the spatial exterior derivative $\mathrm{d}\boldsymbol{\beta}^\flat$:

$$\omega = \left(\mathrm{d}\beta_{\hat{r}} \wedge \boldsymbol{e}^{\hat{r}}\right) + \left(\mathrm{d}\beta_{\hat{\theta}} \wedge \boldsymbol{e}^{\hat{\theta}} + \beta_{\hat{\theta}}\, \mathrm{d}\boldsymbol{e}^{\hat{\theta}}\right) + \left(\mathrm{d}\beta_{\hat{\phi}} \wedge \boldsymbol{e}^{\hat{\phi}} + \beta_{\hat{\phi}}\, \mathrm{d}\boldsymbol{e}^{\hat{\phi}}\right). \tag{28}$$

The spatial exterior derivatives of the shift vector components $\beta_{\hat{r}}, \beta_{\hat{\theta}}, \beta_{\hat{\phi}}$ in the tetrad (vierbein) spherical components are:

$$\mathrm{d}\beta_{\hat{r}} = \frac{\partial \beta_{\hat{r}}}{\partial r}\, \boldsymbol{e}^{\hat{r}} + \frac{1}{r}\, \frac{\partial \beta_{\hat{r}}}{\partial \theta}\, \boldsymbol{e}^{\hat{\theta}} + \frac{1}{r \sin\theta}\, \frac{\partial \beta_{\hat{r}}}{\partial \phi}\, \boldsymbol{e}^{\hat{\phi}}, \tag{29a}$$

$$\mathrm{d}\beta_{\hat{\theta}} = \frac{\partial \beta_{\hat{\theta}}}{\partial r}\, \boldsymbol{e}^{\hat{r}} + \frac{1}{r}\, \frac{\partial \beta_{\hat{\theta}}}{\partial \theta}\, \boldsymbol{e}^{\hat{\theta}} + \frac{1}{r \sin\theta}\, \frac{\partial \beta_{\hat{\theta}}}{\partial \phi}\, \boldsymbol{e}^{\hat{\phi}}, \tag{29b}$$

$$\mathrm{d}\beta_{\hat{\phi}} = \frac{\partial \beta_{\hat{\phi}}}{\partial r}\, \boldsymbol{e}^{\hat{r}} + \frac{1}{r}\, \frac{\partial \beta_{\hat{\phi}}}{\partial \theta}\, \boldsymbol{e}^{\hat{\theta}} + \frac{1}{r \sin\theta}\, \frac{\partial \beta_{\hat{\phi}}}{\partial \phi}\, \boldsymbol{e}^{\hat{\phi}}. \tag{29c}$$

Collecting all terms, we obtain the components of the spatial vorticity 2-form ω in the tetrad basis:

$$\omega = \omega_{\hat{r}\hat{\theta}}\, \boldsymbol{e}^{\hat{r}} \wedge \boldsymbol{e}^{\hat{\theta}} + \omega_{\hat{r}\hat{\phi}}\, \boldsymbol{e}^{\hat{r}} \wedge \boldsymbol{e}^{\hat{\phi}} + \omega_{\hat{\theta}\hat{\phi}}\, \boldsymbol{e}^{\hat{\theta}} \wedge \boldsymbol{e}^{\hat{\phi}}. \tag{30}$$

To obtain the spatial vorticity vector $\boldsymbol{\omega}$, we apply the spatial Hodge star operator to the spatial vorticity 2-form ω:

$$\boldsymbol{\omega} = *\omega = *(\mathrm{d}\boldsymbol{\beta}^\flat). \tag{31}$$

(We use only the spatial (3D) exterior calculus on each slice Σ_t: d denotes the spatial exterior derivative and $*$ the 3D Hodge star. When spacetime derivatives are mentioned, we write d_4 explicitly.)

Therefore the components of the spatial vorticity vector $\omega^{\hat{k}}$ are given by:

$$\omega^{\hat{k}} = (*\omega)^{\hat{k}} = \frac{1}{2}\, \epsilon^{\hat{k}\hat{i}\hat{j}}\, \omega_{\hat{i}\hat{j}}\,, \tag{32}$$

where $\epsilon^{\hat{r}\hat{\theta}\hat{\phi}} = +1$ is the Levi-Civita symbol in the tetrad basis, fully antisymmetric in its indices. Here we have expressed the result in terms of the contravariant components $\beta^{\hat{i}}$, using the numerical equality $\beta_{\hat{i}} = \beta^{\hat{i}}$ for spatial components in the orthonormal

basis. Substituting the expressions for $\omega_{\hat{i}\hat{j}}$, we obtain:

$$\omega^{\hat{r}} = \omega_{\hat{\theta}\hat{\phi}} = \frac{1}{r}\frac{\partial\beta^{\hat{\phi}}}{\partial\theta} - \frac{1}{r\sin\theta}\frac{\partial\beta^{\hat{\theta}}}{\partial\phi} + \frac{\cos\theta}{r\sin\theta}\beta^{\hat{\phi}}, \tag{33}$$

$$\omega^{\hat{\theta}} = \omega_{\hat{\phi}\hat{r}} = \frac{1}{r\sin\theta}\frac{\partial\beta^{\hat{r}}}{\partial\phi} - \frac{\partial\beta^{\hat{\phi}}}{\partial r} - \frac{1}{r}\beta^{\hat{\phi}}, \tag{34}$$

$$\omega^{\hat{\phi}} = \omega_{\hat{r}\hat{\theta}} = \frac{\partial\beta^{\hat{\theta}}}{\partial r} + \frac{\beta^{\hat{\theta}}}{r} - \frac{1}{r}\frac{\partial\beta^{\hat{r}}}{\partial\theta}. \tag{35}$$

As expected, these tetrad components correspond to the standard physical vorticity components in spherical coordinates as used in fluid mechanics [31, 32].

We seek an irrotational shift vector field $\boldsymbol{\beta}$ exhibiting continuous rotational symmetry about the x-axis (the axis of travel). This axial symmetry implies that the components of $\boldsymbol{\beta}$ in the spherical orthonormal basis must be independent of the azimuthal angle ϕ ($\partial/\partial\phi = 0$) and that the azimuthal component itself must vanish ($\beta^{\hat{\phi}} = 0$). We therefore adopt the following ansatz for the tetrad components, separating variables in (t, r, θ):

$$\begin{pmatrix} \beta^{\hat{r}} \\ \beta^{\hat{\theta}} \\ \beta^{\hat{\phi}} \end{pmatrix} = \begin{pmatrix} -v(t)\, f(r)\, \cos\theta \\ v(t)\, g(r)\, \sin\theta \\ 0 \end{pmatrix}. \tag{36}$$

Here, $v(t)$ is the velocity along the x-axis, while $f(r)$ and $g(r)$ define the radial profile. The relative signs ensure consistency with motion along the $+x$ direction (as discussed for the Alcubierre case in Sec. 3.2).

Substituting this axially symmetric form ($\partial/\partial\phi = 0$, $\beta^{\hat{\phi}} = 0$) into the general expressions for the vorticity components (33)-(34), we immediately find $\omega^{\hat{r}} = 0$ and $\omega^{\hat{\theta}} = 0$. The kinematic irrotationality condition, $\boldsymbol{\omega} = 0$, thus reduces solely to requiring the azimuthal component $\omega^{\hat{\phi}}$ to vanish. From (35), this condition becomes:

$$\omega^{\hat{\phi}} = \frac{\partial\beta^{\hat{\theta}}}{\partial r} + \frac{\beta^{\hat{\theta}}}{r} - \frac{1}{r}\frac{\partial\beta^{\hat{r}}}{\partial\theta} = 0. \tag{37}$$

By Frobenius's theorem, this condition is equivalent to the spatial flow being surface-orthogonal *on each slice*: since $\mathrm{d}\beta^\flat = 0$ on Σ_t, there exists a scalar potential Φ with $\beta^\flat = -\mathrm{d}\Phi$, so $\boldsymbol{\beta}$ is orthogonal to the 2-surfaces $\Phi = \text{const}$ within the slice (this is distinct from spacetime vorticity). This requirement leads to a differential equation for the tetrad components $\beta^{\hat{i}}$, which can be solved to construct the desired warp-drive spacetime.

Substituting (36) into (37), we obtain:

$$v(t)\sin\theta\left(\frac{\partial g(r)}{\partial r} + \frac{g(r)}{r} - \frac{f(r)}{r}\right) = 0. \tag{38}$$

Since the prefactor $v(t)\sin\theta \neq 0$ for $\theta \neq 0, \pi$; the irrotational condition $\omega^{\hat{\phi}} = 0$ reduces to a linear first-order ordinary differential equation (ODE) $\frac{\partial g(r)}{\partial r} + \frac{g(r)}{r} = \frac{f(r)}{r}$ relating $g(r)$ and $f(r)$. This linear first-order ODE for $g(r)$ can be solved using the integrating factor $\exp\left(\int \frac{1}{r}\, dr\right) = r$. Multiplying by r yields $r\frac{dg}{dr} + g(r) = f(r)$, or $\frac{d}{dr}(rg(r)) = f(r)$. Integrating with respect to r gives $rg(r) = \int f(r)\, dr + C$, leading to:

$$g(r) = \frac{1}{r}\left(\int f(r)\, dr + C\right). \tag{39}$$

This expression provides $g(r)$ in terms of $f(r)$ and the integration constant C, to be determined from boundary conditions.

In the spatially divergence-free warp drive, the shift vector component in the polar angle direction involves the *derivative* (21) of the form function $f(r)$ with respect to r. Conversely, in the irrotational warp drive, this component involves the *integral* (39) of the form function $f(r)$ with respect to r. The choice of $f(r)$ is guided by physical considerations, particularly the desired boundary conditions for the shift vector. Adopting the same convention as for the Natário drive regarding the observer perspective (Sec. 3.3, observer stationary at $r = 0$, distant stars recede at $-v$), we require the form function $f(r)$ to satisfy $\lim_{r\to\infty} f(r) = 1$ and $\lim_{r\to 0} f(r) = 0$. As in Sec. 3.3, we achieve this using a definition based on Alcubierre's function (14):

$$f(r) := 1 - f_{\text{Alc}}(r). \tag{40}$$

With this form function it follows from (39) and (40), that evaluating the integral analytically yields $g(r)$ as follows:

$$\begin{aligned} g(r) = \frac{\operatorname{csch}\left(\frac{\rho\sigma}{2}\right)\operatorname{sech}\left(\frac{\rho\sigma}{2}\right)}{4\,r\,\sigma}\Bigg[&2r\sigma\sinh(\rho\sigma) \\ &+ \cosh(\rho\sigma)\left(\ln\Big(\cosh(\sigma(r-\rho))\Big) - \ln\Big(\cosh(\sigma(r+\rho))\Big)\right)\Bigg] + \frac{C}{r}. \end{aligned} \tag{41}$$

The boundary condition $\lim_{r\to 0} g(r) = 0$ requires the integration constant C to be set to zero, i.e., $C = 0$. This boundary condition determines $g(r)$ to be:

$$g(r) = \frac{2r\sigma\sinh(\rho\sigma) + \cosh(\rho\sigma)\left(\ln\left(\frac{\cosh(\sigma(r-\rho))}{\cosh(\sigma(r+\rho))}\right)\right)}{4\,r\,\sigma\sinh\left(\frac{\rho\sigma}{2}\right)\cosh\left(\frac{\rho\sigma}{2}\right)}. \tag{42}$$

This function $g(r)$, given by (42) with $C = 0$, by construction ensures zero kinematic vorticity when used with $f(r) = 1 - f_{\text{Alc}}(r)$. This $g(r)$ satisfies $\lim_{r\to 0} g(r) = 0$ and $\lim_{r\to\infty} g(r) = 1$. A short, rigorous proof of the $r \to 0$ behavior for $f(r) = 1 - f_{\text{Alc}}(r)$ is provided in Appendix B. Consequently, the full shift vector defined by $f(r)$ and $g(r)$ in (36) satisfies the desired asymptotic conditions shown in the Natário

rows of Table E7 (Appendix E, at $\theta = 0$ and $\theta = \pi/2$). These boundary conditions are satisfied due to the properties of the hyperbolic functions and the chosen forms of $f(r)$ and $g(r)$ provided that: $\sigma > 0$, $(v(t),\, \rho) \in \mathbb{R}$.

Scalar potential

The kinematic irrotationality condition $\mathrm{d}\boldsymbol{\beta}^\flat = 0$ ensures that the shift covector field can, on topologically simple spatial slices, be derived globally from a scalar potential $\Phi(r, \theta, t)$. Adopting the convention $\boldsymbol{\beta}^\flat = -\mathrm{d}\Phi$ inherently satisfies the irrotationality condition, as $\mathrm{d}\boldsymbol{\beta}^\flat = -\mathrm{d}^2\Phi = 0$. For the specific potential consistent with the integration constant $C = 0$ and the relationship $\frac{d}{dr}(rg(r)) = f(r)$,

$$\Phi(r, \theta, t) = v(t)\big(r\, g(r)\big) \cos\theta, \tag{43}$$

where $g(r)$ is given by (42), we compute the covariant components of $\boldsymbol{\beta}^\flat = \beta_{\hat{\alpha}} \boldsymbol{e}^{\hat{\alpha}}$ in the orthonormal tetrad basis $\{\boldsymbol{e}^{\hat{\alpha}}\}$. The components of the exterior derivative $\mathrm{d}\Phi$ in this basis are given by the projections of the gradient onto the tetrad vectors, leading to $\beta_{\hat{\mu}} = -e^{\nu}{}_{\hat{\mu}} \partial_\nu \Phi$. For the standard spherical orthonormal tetrad where $\boldsymbol{e}^{\hat{r}} = \mathrm{d}r$ and $\boldsymbol{e}^{\hat{\theta}} = r\,\mathrm{d}\theta$, this yields:

$$\beta_{\hat{r}} = -e^{\nu}{}_{\hat{r}} \partial_\nu \Phi = -\partial_r \Phi = -v(t) f(r) \cos\theta, \tag{44}$$

$$\beta_{\hat{\theta}} = -e^{\nu}{}_{\hat{\theta}} \partial_\nu \Phi = -\frac{1}{r} \partial_\theta \Phi = v(t) g(r) \sin\theta, \tag{45}$$

$$\beta_{\hat{\phi}} = -e^{\nu}{}_{\hat{\phi}} \partial_\nu \Phi = -\frac{1}{r \sin\theta} \partial_\phi \Phi = 0. \tag{46}$$

Recalling that $\beta^{\hat{i}} = \eta^{\hat{i}\hat{j}} \beta_{\hat{j}} = \delta^{\hat{i}\hat{j}} \beta_{\hat{j}} = \beta_{\hat{i}}$ for the spatial components $(i, j \in \{r, \theta, \phi\})$ in the chosen orthonormal tetrad, these derived components (44)–(45) correspond precisely to the ansatz components $\beta^{\hat{r}}, \beta^{\hat{\theta}}, \beta^{\hat{\phi}}$ defined in (36). As previously discussed, however, no classical thermodynamic or fluid conservation interpretation attaches to this potential Φ in the context of warp-drive geometry.

Proposition 2 (Hawking–Ellis classification of the irrotational drive)**.** ***Assumptions:*** $\alpha \equiv 1$, *flat* and static *spatial slices* $({}^{(3)}R_{ij} = 0,\ \partial_t \gamma_{ij} = 0)$*, and an irrotational shift* $\boldsymbol{\beta}^\flat = -\mathrm{d}\Phi$*. Then the stress–energy tensor of the derived irrotational warp–drive spacetime is Hawking–Ellis Type I at every point.*

Proof. With $\alpha = 1$ and a flat static slice ($\gamma_{ij} \simeq \mathbb{R}^3$), the extrinsic curvature is the spatial Hessian $K_{\hat{i}\hat{j}} = -D_{\hat{i}} D_{\hat{j}} \Phi$. The momentum constraint $G_{\hat{0}\hat{i}} \propto D_{\hat{k}}\big(K^{\hat{k}}{}_{\hat{i}} - \delta^{\hat{k}}_{\hat{i}} K\big) = -D_{\hat{k}}\big(D^{\hat{k}} D_{\hat{i}} \Phi - \delta^{\hat{k}}_{\hat{i}} D_{\hat{m}} D^{\hat{m}} \Phi\big)$ vanishes identically because $[D_{\hat{k}}, D_{\hat{i}}] = 0$ on the Riemann-flat slice. Hence $G^{\hat{\alpha}}{}_{\hat{\beta}}$ is block-diagonal in the orthonormal tetrad adapted to the Eulerian normal n^μ (in the coordinate basis $\{\partial_t, \partial_i\}$), with $n^\mu = (1, \beta^i)$ for $\alpha = 1$, and coframe $\omega^{\hat{0}} = dt$, $\omega^{\hat{i}} = dx^i - \beta^i dt$; equivalently, $\partial_t = \alpha\, n^\mu \partial_\mu - \beta^i \partial_i$ so: $\partial_t = n - \beta^i \partial_i$. For flat, time–independent spatial slices $({}^{(3)}R_{ij} = 0,\ \partial_t \gamma_{ij} = 0)$:

$$G^{\hat{\alpha}}{}_{\hat{\beta}} = \mathrm{diag}\big(G^{\hat{0}}{}_{\hat{0}}, G^{\hat{i}}{}_{\hat{j}}\big), \qquad G^{\hat{i}}{}_{\hat{j}} = G^{\hat{j}}{}_{\hat{i}}\,. \tag{47}$$

The spatial block is real symmetric, so its three eigenvalues $\lambda_{G(k)}$ $(k = 1, 2, 3)$ are real; $\lambda_{G(0)} = G^{\hat{0}}{}_{\hat{0}}$ is also real. Because $e_{\hat{0}}$ is timelike and is the eigenvector for $\lambda_{G(0)}$, the set $\{\lambda_{G(\alpha)}\}_{\alpha=0}^{3}$ comprises four real eigenvalues with a timelike eigenvector. By the Hawking–Ellis scheme this is precisely Type I [15, 16]. □

Mechanism for Peak Reduction

The observed reduction in stress–energy extrema is a direct consequence of kinematic irrotationality and its algebraic implications. In warp drives that permit vorticity (e.g., Alcubierre and Natário [1, 2]), a nonzero vorticity ($\mathrm{d}\boldsymbol{\beta}^\flat \neq 0$) typically yields a nonzero momentum density $G_{\hat{0}\hat{i}}$, introducing energy–flux terms that can lead to patches of Hawking–Ellis Type IV structure [21, 33].

By contrast, under the present assumptions ($\alpha \equiv 1$, flat and time–independent spatial slices, and $\mathrm{d}\boldsymbol{\beta}^\flat = 0$), the 3+1 momentum constraint enforces vanishing momentum density:

$$G_{\hat{0}\hat{i}} = 0, \tag{48}$$

which block–diagonalizes the mixed Einstein tensor in the Eulerian orthonormal tetrad:

$$G^{\hat{\mu}}{}_{\hat{\nu}} = \mathrm{diag}\big(G^{\hat{0}}{}_{\hat{0}},\ G^{\hat{i}}{}_{\hat{j}}\big). \tag{49}$$

This guarantees a global Hawking–Ellis Type I structure and is the algebraic origin of the improved energy–condition profile.

This causal link is not merely theoretical; it is confirmed by a rigorous numerical ablation (Appendix D). In that test, the baseline smoothing profile $g(r)$ and numerical tolerances are held fixed while a tunable vortical component is introduced, isolating vorticity as the sole variable. The results (Table D6) show a catastrophic degradation of the energy metrics: the negative–energy magnitude E_- increases by a factor exceeding 500, and the crucial E_+/E_- balance collapses. This demonstrates that the favorable energy properties of the baseline model are a direct consequence of its irrotational, curl–free kinematics, not an artifact of profile smoothing.

Comparative summary of warp-drive models

Table 3 provides a comparative summary of the Alcubierre [1], Natário zero-expansion [2], and kinematically irrotational warp drives, based on the orthonormal tetrad components derived in Secs. 3.1, 3.2, 3.3 and 3.4.2 (assuming lapse function $\alpha = 1$, flat spatial slices, axial symmetry with x-axis polar). The table contrasts the defining kinematic characteristic imposed on the shift vector $\boldsymbol{\beta}$, the resulting tetrad components $\beta^{\hat{i}}$, the utilized radial form functions ($f(r)$, $g(r) = \frac{1}{r}(\int f(r)\,\mathrm{d}r)$ derived from $f_{\mathrm{Alc}}(r)$ (14)), and the inferred observer perspective. This highlights how the specific constraints ($\mathrm{div}\,\boldsymbol{\beta} = 0$ for Natário, $\boldsymbol{\omega} = *\mathrm{d}\boldsymbol{\beta}^\flat = 0$ for the irrotational model) manifest in the shift vector components relative to the original Alcubierre spacetime.

Table 3: Warp-drive models

Model	**Alcubierre [1]**	**Natário (No Exp.) [2]**	**Kinem. irrotational**
Charact.	$\boldsymbol{\beta}^\flat = v f_{\mathrm{Alc}} \mathrm{d}x$ (12)	$\operatorname{div} \boldsymbol{\beta} = 0$ (15)	$\boldsymbol{\omega} = *(\mathrm{d}\boldsymbol{\beta}^\flat) = 0$ (31)
Shift Vr. ($\beta^{\hat{i}}$ Tetrad)	from (13): $\beta^{\hat{r}} = v f \cos\theta$ $\beta^{\hat{\theta}} = -v\, f \sin\theta$ $\beta^{\hat{\phi}} = 0$	from (21): $\beta^{\hat{r}} = -v f \cos\theta$ $\beta^{\hat{\theta}} = v(f + \frac{r}{2} f') \sin\theta$ $\beta^{\hat{\phi}} = 0$	from (36) and (39): $\beta^{\hat{r}} = -v f \cos\theta$ $\beta^{\hat{\theta}} = v \frac{1}{r} (\int f \, \mathrm{d}r) \sin\theta$ $\beta^{\hat{\phi}} = 0$
Form Fn.	$f = f_{\mathrm{Alc}}$ (14)	$f = 1 - f_{\mathrm{Alc}}$ (22)	$f = 1 - f_{\mathrm{Alc}}$ (40)
Observer	Rest $r \to \infty$; Bubble $v(t)$	Rest $r = 0$; Stars $-v(t)$	Rest $r = 0$; Stars $-v(t)$

4 Interpretation and visualization of Einstein-tensor eigenvalues

This section connects the algebraic framework developed above with the numerical results for the *Alcubierre*, *irrotational* and *Natário* warp-drive spacetimes. Subsection 4.1 reviews the Cartan construction of $G^{\hat{\alpha}}{}_{\hat{\beta}}$, applies the Hawking–Ellis classification (emphasizing Types I and IV), and presents the eigensolver, tolerance schemes and validation checks that lead to the physically-motivated ordering of the eigenpairs.

Section 4.2 analyzes the plotted eigen-data, maps regions where the Einstein tensor is Hawking–Ellis Type I (fully diagonalizable) or Type IV (complex–conjugate null pair), and quantifies violations of both the null and weak energy conditions. It also computes the scalar invariant $G = -R = \sum_{\alpha=0}^{3} \lambda_{G(\alpha)}$ which enters the trace energy condition, thereby enabling a direct comparison of the stress–energy requirements for the three warp-drive spacetimes.

Numerical values of the mixed Einstein tensor $G^{\hat{\mu}}{}_{\hat{\nu}}$, expressed in an orthonormal tetrad, are computed in Wolfram *Mathematica*® [34] at 50-digit precision (unless stated otherwise). We examine a warp bubble moving at light-speed with the parameters $\rho = 5\,[\mathrm{m}]$ and $\sigma = 4\,[\mathrm{m}^{-1}]$. These choices reproduce those of Refs. [35, 36], which employed a curvilinear-coordinate tensor formalism; by using the same parameters in our Cartan orthonormal-tetrad approach (with spin coefficients), we can directly compare the computed $G^{\hat{\mu}}{}_{\hat{\nu}}$ for the Alcubierre and Natário drives, thus thoroughly validating our high-precision numerical implementation. To economize space, we restrict attention to the light-speed case here; superluminal warp-bubble velocities produce analogous qualitative behavior (as follows from the analytic expressions) and have been numerically detailed for the Alcubierre and Natário drives in Ref. [36]. All figures sample the radial band $r \in \left[\rho - 3/\sigma,\ \rho + 3/\sigma\right]$, which contains the bubble walls where the form function $f(r)$ departs from its asymptotic limits $f \to 1$ and $f \to 0$.
Methods note: All eigenvalue computations used 50-digit precision with symmetry, trace, and eigen-sum checks; results were insensitive to tolerance variations over two orders of magnitude.

Global energy metrics

We summarize the slice–integrated measures defined in Appendix C: the negative/-positive volumes $V_\pm$ and energy magnitudes $E_\pm$. These are integrals over a spatial

slice Σ_t and hence are foliation-dependent; they are invariant under spatial coordinate changes and tetrad rotations on Σ_t, but not under changes of slicing. They are computed with the disjoint masks of Eqs. (C.6)–(C.7), using the axially reduced volume element (C.8), and the definitions in Eqs. (C.11)–(C.12). Headline values for the fiducial $(\rho, \sigma, v/c) = (5\ [\mathrm{m}],\ 4\ [\mathrm{m}^{-1}],\ 1)$ on the baseline window $R_{\mathrm{integ}} = 12\rho$ appear in Tables C2 and C3; localization diagnostics are reported in Tables C4–C5 and the meridional map in Fig. C1. These global measures complement the local extrema (e.g. $\min \varrho_p$) and quantify the spatial extent and integrated magnitude of NEC/WEC–violating regions. A compact snapshot is given in Table 4; for far–field completion using a two–point $1/R$ tail model and corresponding totals, see Sec. C and Table C1, which drive the net proper energy to $|E_{\mathrm{net}}|/E_{\mathrm{abs}} = 0.04\%$ (i.e. a 4×10^{-4} fractional level), consistent with zero.

Table 4: Local and global *energy* measures for the irrotational warp drive at the fiducial $(\rho, \sigma, v/c)$. Baseline values are on the window $R_{\mathrm{integ}} = 12\rho$; tail–corrected values use a two–point $1/R$ extrapolation to $R \to \infty$. Numbers are reported with 3 s.f. .

Case	E_- [J]	E_+ [J]	E_+/E_-
Baseline (12ρ)	1.21×10^{44}	1.29×10^{44}	1.07
Tail–corrected ($R \to \infty$, $1/R$)	1.33×10^{44}	1.33×10^{44}	1.00

Local extrema and baseline volumes (at $R_{\mathrm{integ}} = 12\rho$)				
$\max \varrho_p$ [J m^{-3}]	$\min \varrho_p$ [J m^{-3}]	V_- [m^3]	V_+ [m^3]	V_0 [m^3]
2.23×10^{42}	-1.29×10^{41}	6.38×10^5	2.34×10^5	3.40×10^4

Limitations. All Type I and peak-reduction results reported here assume $\alpha \equiv 1$ and flat, *time-independent* spatial slices. Relaxing these conditions (e.g., allowing $\nabla\alpha \neq 0$ or evolving γ_{ij}) can reintroduce momentum density $G_{\hat{0}\hat{i}}$ and spoil global Type I behavior.

4.1 Algebraic classification, interpretation, and numerical protocol for $G^\mu{}_\nu$

The Einstein tensor $G^\mu{}_\nu = g^{\mu\alpha} G_{\alpha\nu}$ shares the same algebraic structure as the stress–energy tensor $T^\mu{}_\nu$. With the cosmological constant set to zero $\Lambda = 0$, Einstein's field equations $G^\mu{}_\nu = \kappa T^\mu{}_\nu$ imply a pointwise proportionality between these two tensors, so the Hawking–Ellis (Segre–Plebański) classification [19, 37–39] can be applied directly to $G^\mu{}_\nu$. In practice this means that computing the eigenvalues and causal character of $G^\mu{}_\nu$ directly reveals the type of stress-energy tensor or effective source required. This is particularly useful in warp-drive metrics, where one specifies $g_{\mu\nu}$ *a priori* and then reads off the implied energy–momentum content rather than solving for it.

4.1.1 Einstein tensor via Cartan formalism

The mixed components $G^{\hat{\alpha}}{}_{\hat{\beta}}$ are computed entirely in Cartan's tetrad language, using Wald's sign conventions [28] and avoiding Christoffel symbols [29]. We choose an orthonormal coframe $\{\omega^{\hat{\alpha}}\} \equiv \{e^{\hat{\alpha}}{}_{\mu}\,\mathrm{d}x^{\mu}\}$ with $g_{\mu\nu} = \eta_{\hat{\alpha}\hat{\beta}}\, e^{\hat{\alpha}}{}_{\mu}\, e^{\hat{\beta}}{}_{\nu}$, $\eta_{\hat{\alpha}\hat{\beta}} = \mathrm{diag}(-,+,+,+)$. The dual frame vectors $\mathbf{e}_{\hat{\alpha}} = e_{\hat{\alpha}}{}^{\mu}\partial_{\mu}$ satisfy the usual completeness relations.

From the first Cartan structure equation (zero torsion, $T^{\hat{\alpha}} = 0$):

$$\mathrm{d}\,\omega^{\hat{\alpha}} + \omega^{\hat{\alpha}}{}_{\hat{\beta}} \wedge \omega^{\hat{\beta}} = 0, \tag{50}$$

one reads off the antisymmetric spin-connection 1-forms $\omega^{\hat{\alpha}}{}_{\hat{\beta}} = -\omega_{\hat{\beta}}{}^{\hat{\alpha}}$.

The second Cartan structure equation defines the curvature 2-forms:

$$\Omega^{\hat{\alpha}}{}_{\hat{\beta}} = \mathrm{d}\,\omega^{\hat{\alpha}}{}_{\hat{\beta}} + \omega^{\hat{\alpha}}{}_{\hat{\gamma}} \wedge \omega^{\hat{\gamma}}{}_{\hat{\beta}} = \tfrac{1}{2} R^{\hat{\alpha}}{}_{\hat{\beta}\hat{\gamma}\hat{\delta}}\, \omega^{\hat{\gamma}} \wedge \omega^{\hat{\delta}}. \tag{51}$$

Contracting $R^{\hat{\alpha}}{}_{\hat{\beta}\hat{\gamma}\hat{\delta}}$ yields $R_{\hat{\alpha}\hat{\beta}} = R^{\hat{\gamma}}{}_{\hat{\alpha}\hat{\gamma}\hat{\beta}}$, $\quad R = \eta^{\hat{\alpha}\hat{\beta}} R_{\hat{\alpha}\hat{\beta}}$. Finally, the Einstein tensor is $G_{\hat{\alpha}\hat{\beta}} = R_{\hat{\alpha}\hat{\beta}} - \frac{1}{2}\, R\, \eta_{\hat{\alpha}\hat{\beta}}$, $\quad G^{\hat{\alpha}}{}_{\hat{\beta}} = \eta^{\hat{\alpha}\hat{\gamma}} G_{\hat{\gamma}\hat{\beta}}$, and we feed this 4×4 matrix to the eigenanalysis.

4.1.2 Hawking–Ellis Type IV: geometry and physical meaning

Type IV occurs when the mixed Einstein tensor $G^{\hat{\alpha}}{}_{\hat{\beta}}$ possesses one complex–conjugate eigen-pair and two distinct real spacelike eigenvalues; consequently, no real causal (timelike) eigenvector exists. Segre [39, §20, p. 330] records this four-simple-divisor spectrum as $[1\,1\,1\,1]$ when viewed over $\mathbb{C}$. Petrov adopts the same bracket [40, §47, pp. 324–325], while Stephani et al. [37, Table 5.1] write it as $[1\,1,\, Z\bar{Z}]$. Following Hall [41, Table 7.1] we use the label $[z\,\bar{z}\,11]$.

Let the complex eigenvalues be

$$\Lambda_G = \xi + i\mu, \quad \bar{\Lambda}_G = \xi - i\mu, \quad \lambda_2, \lambda_3 \in \mathbb{R}, \quad \lambda_2 \neq \lambda_3, \quad \mu \neq 0. \tag{52}$$

with eigenvectors z^{μ} and $\bar{z}^{\mu}$. Because $G^{\hat{\alpha}}{}_{\hat{\beta}}$ is self-adjoint, $g(G(z), \bar{z}) = g(z, G(\bar{z}))$. Substituting $G(z) = \Lambda_G z$ and $G(\bar{z}) = \bar{\Lambda}_G \bar{z}$, this gives $(\Lambda_G - \bar{\Lambda}_G)\, g(z, \bar{z}) = 0$. Since $\Lambda_G \neq \bar{\Lambda}_G$, we have $g(z, \bar{z}) = 0$.

Null two-plane.

Write $z^{\mu} = x^{\mu} + i\, y^{\mu}$. Then $g(x,x) + g(y,y) = 0$. Without loss of generality take x^{μ} timelike and y^{μ} spacelike, orthogonal; the real plane $\mathscr{N} = \mathrm{span}\{x^{\mu}, y^{\mu}\}$ is Lorentzian and contains two independent null directions. Every genuine Type-IV tensor shares this intrinsic null two-plane; no additional branch with $g(z, \bar{z}) \neq 0$ is algebraically possible. This impossibility traces directly to the g-self-adjointness of $G^{\hat{\alpha}}{}_{\hat{\beta}}$.

Canonical 2×2 ***block.***

Let $\{t^\mu, s^\mu\} \subset \mathscr{N}$ be an orthonormal basis of the invariant null two–plane, with $g(t,t) = -1,\ g(s,s) = +1,\ g(t,s) = 0$. Then

$$G^\mu{}_\nu\, t^\nu = \xi\, t^\mu - \mu\, s^\mu, \tag{53a}$$

$$G^\mu{}_\nu\, s^\nu = \mu\, t^\mu + \xi\, s^\mu, \tag{53b}$$

so on the $t-s$ plane the mixed Einstein tensor is represented by

$$\begin{pmatrix} \xi & \mu \\ -\mu & \xi \end{pmatrix}.$$

Canonical forms across the literature. Our mixed 2×2 Type-IV block matches the canonical forms in [19, 21, 33, 40] after the appropriate index-position conversion (covariant or contravariant $\leftrightarrow$ mixed) and signature bookkeeping; explicit parameter maps are given in App. F.

Physical measurables.

With Einstein's coupling constant κ, an observer whose four–velocity is t^μ (aligned with the null plane) measures

$$\varrho_{\text{obs}} = -\xi/\kappa, \quad p_s = \xi/\kappa, \quad q_s = \mu/\kappa,$$

where q_s is an unavoidable energy- (or momentum-) flux in the s^μ direction. As shown in Prop. 3, the quantities ξ and $|\mu|$ are Lorentz scalars, but ϱ_{obs} itself is frame-dependent because no real causal eigenvector exists.

Proposition 3 (Invariant magnitudes of ξ and μ)**.** *Let $G^{\hat\alpha}{}_{\hat\beta}$ at a spacetime point be Hawking–Ellis Type IV, with complex eigenvalues $\Lambda_G = \xi \pm i\mu$. Then ξ and $|\mu|$ are Lorentz scalars (only the sign of μ can flip). Consequently $|q_s| = |\mu|/\kappa$ is invariant, and for observers whose 4-velocities lie in the null two-plane $\mathscr{N}$ one has $\varrho_{obs} = -\xi/\kappa$, also invariant within that class.*

Proof. In an orthonormal tetrad adapted to $\mathscr{N}$,

$$B \;=\; G^{\hat\alpha}{}_{\hat\beta}\big|_{\mathscr{N}} = \begin{pmatrix} \xi & \mu \\ -\mu & \xi \end{pmatrix}, \qquad \mu \neq 0.$$

Any Lorentz transformation that preserves $\mathscr{N}$ acts by real similarity on B, hence

$$\operatorname{tr} B = 2\xi, \qquad \det B = \xi^2 + \mu^2$$

are invariants. Since $\mu^2 = \det B - \frac{1}{4}(\operatorname{tr} B)^2$ is invariant, only $|\mu|$ is well defined (orientation reversals send $\mu \to -\mu$). Dividing by κ gives the stated invariants for $|q_s|$ and, for null-plane observers, ϱ_{obs}. $\square$

Although $\xi = \frac{1}{2}\,\mathrm{tr}\,B$ is a true scalar, it is *not* a proper energy density: no real timelike eigenvector exists. The energy density measured by an arbitrary observer $t^{\hat{\gamma}}$ is

$$\varrho_{\rm obs}[t^{\hat{\gamma}}] \;=\; \frac{1}{\kappa}\,G_{\hat{\alpha}\hat{\beta}}\,t^{\hat{\alpha}}t^{\hat{\beta}},$$

and varies with the frame. Only observers whose four-velocities lie in $\mathscr{N}$ obtain the invariant value $\varrho_{\rm obs} = -\xi/\kappa$. By contrast, in a Type I region the unique timelike eigenvalue yields the proper (frame-invariant) energy density $\varrho_p = -\lambda_{G(0)}/\kappa$.

Null–energy–condition violation

Because the energy–flux parameter μ attached to the complex eigenvalues is non-zero ($\mu \neq 0$), *every* Hawking–Ellis Type IV tensor violates the null energy condition, $G_{\mu\nu}K^\mu K^\nu \geq 0$, for a suitable null vector K^μ. Write the complex eigenvector as $z^\mu = x^\mu + iy^\mu$ with $g(x,x) = -g(y,y)$ and $g(x,y) = 0$; the invariant two–plane $\mathscr{N} = \mathrm{span}\{x,y\}$ then contains the two null directions $K^\mu_\pm = x^\mu \pm y^\mu$. A direct calculation gives

$$G_{\mu\nu}K^\mu_\pm K^\nu_\pm = \mp\,2\,|\mu|\;|g(x,x)|,$$

so one choice of sign is strictly negative, establishing NEC violation.

Structural stability

Type IV is *structurally unstable*. Its defining algebraic constraint $g(z,\bar{z}) = 0$ is an equality that is generically spoiled by an arbitrarily small perturbation of the tensor, causing the eigen-system to move into another Hawking–Ellis class (typically Type I, or, at special degeneracy points, Type II/III). By contrast, Type I—with four distinct real eigenvalues—is an open, stable class. (Types II and III are likewise unstable because they rely on eigenvalue degeneracies.)

4.1.3 Numerical protocol and validation

We analyze the mixed Einstein tensor $G^{\hat{\alpha}}{}_{\hat{\beta}} = G^\mu{}_\nu\, e^{\hat{\alpha}}{}_\mu\, e^\nu{}_{\hat{\beta}}$ numerically in Wolfram *Mathematica*® [34]. The symbolic expression $G^{\hat{\alpha}}{}_{\hat{\beta}}(r,\theta;\rho,\sigma,v)$ is evaluated at selected coordinates (r_0,θ_0) and warp bubble parameters, (ρ_0,σ_0,v_0), to obtain the numeric 4×4 real matrix

$$M = G^{\hat{\alpha}}{}_{\hat{\beta}}(r_0,\theta_0;\rho_0,\sigma_0,v_0) \in \mathbb{R}^{4\times 4}, \tag{54}$$

All numerical computations are performed using $p = 50$-digit precision by default. The raw eigensystem $\{\lambda_I, \boldsymbol{v}_I\}_{I=1..4}$ of M is computed using `Eigensystem[]`.

Phase fixing of eigenvectors

Each raw eigenvector $\boldsymbol{v}_I$ is then multiplied by a unit-modulus factor $e^{\mathrm{i}\psi} \in U(1)$ (a *global phase*). The phase ψ is chosen such that the time-like component $v^{\hat{0}}$ of the resulting vector becomes real and non-negative; if $|v^{\hat{0}}| < \epsilon_{\rm chop} = 10^{-18}$, ψ is instead chosen to make the largest-magnitude component real and non-negative. Because this rotation by $e^{\mathrm{i}\psi}$ leaves scalar products like $\eta_{\hat{\alpha}\hat{\beta}}v^{\hat{\alpha}}\bar{v}^{\hat{\beta}}$ unchanged, this phase-fixing does not affect any physical result derived from such products (e.g., eigenvector norms) but ensures numerical reproducibility. The set of phase-fixed eigenvectors is denoted $\{\boldsymbol{v}'_I\}_{I=1..4}$.

Classification procedure

From the raw eigensystem $\{\lambda_I, \boldsymbol{v}'_I\}$ we decide the Hawking–Ellis type; N_C counts non-real eigenvalues.

(I) **Type I**: $N_C = 0$.

(II) **Type II**: at least one *real* null eigenvector (Jordan vs. diagonal via $\mathrm{rank}(M - \lambda I)$).

(III) **Type IV**: $N_C = 2$ *and* the complex eigen-vectors satisfy $g(z, \bar{z}) = 0$ (null two-plane).

Tests are attempted in the order I $\to$ II $\to$ IV.

Numerical tolerances

$$\begin{aligned} |\operatorname{Im}\lambda_I| > \epsilon_{\text{imag}} = 10^{-18} \quad &\Rightarrow \lambda_I \text{ is complex,} \\ |\lambda_I - \lambda_J| < \epsilon_{\text{degen}} = 10^{-10} \quad &\Rightarrow \text{near-degeneracy,} \\ |\eta_{\hat{\alpha}\hat{\beta}} v'^{\hat{\alpha}} \bar{v}'^{\hat{\beta}}| < \epsilon_{\text{null}} = 10^{-12} \quad &\Rightarrow \boldsymbol{v}' \text{ is null.} \end{aligned} \tag{55}$$

Initial numerical ordering

After `Eigensystem[]` the raw list $\{\lambda_I, \boldsymbol{v}'_{(I)}\}_{I=1}^4$ is sorted *numerically* according to the Hawking–Ellis type assigned by the classifier:

- **Type I or II**: all eigen-values are real; order by descending $\operatorname{Re}\lambda$.
- **Type IV**: complex conjugate pair first ($\operatorname{Im}\lambda > 0$ placed before $\operatorname{Im}\lambda < 0$), then the two real roots in descending $\operatorname{Re}\lambda$.

This step yields a deterministic but non-physical indexing.

Physical reordering

The list is then mapped to the physically ordered eigensystem $\{\lambda_{G(\alpha)}, \boldsymbol{v}_{G(\alpha)}\}_{\alpha=0}^3$:

- **Type I or II**: $\boldsymbol{v}_{G(0)}$ is the unique causal eigenvector—timelike for I ($g < -\epsilon_{\text{null}}$), *numerically* null for II ($|g| < \epsilon_{\text{null}}$).(The absolute-value tolerance accounts for round-off error.) The remaining spacelike vectors occupy $G(1), G(2), G(3)$ via a two-key sort: (i) dominant spatial component, ascending (r, θ, ϕ); (ii) tie-breaker $\operatorname{Re}\lambda$, descending.
- **Type IV**: the complex–conjugate pair is placed in $G(0), G(1)$ (the root with $\operatorname{Im}\lambda > 0$ goes to $G(0)$, the one with $\operatorname{Im}\lambda < 0$ to $G(1)$); the two real spacelike eigenvectors then fill $G(2), G(3)$ using the same two-key rule.

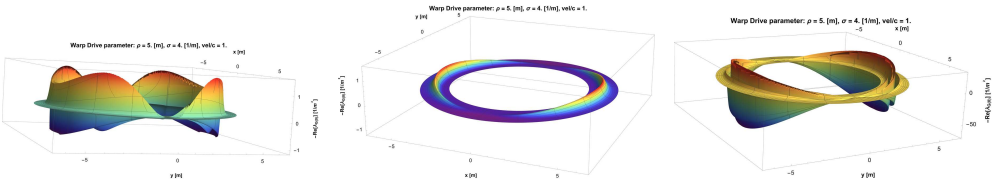
This two-stage procedure ensures that every slot label $G(0)$–$G(3)$ refers to the same physical quantity at every grid point, regardless of the raw eigensolver order. If no unique causal eigenvector is found, the original numerical order is retained.

Convention for the Physically Ordered Spectrum

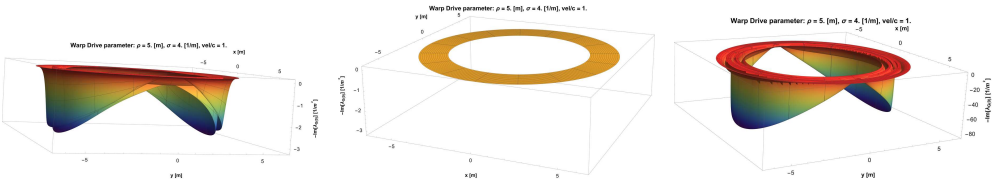
The final output of our numerical pipeline is the physically ordered set of eigenvalues $\{\lambda_{G(0)}, \lambda_{G(1)}, \lambda_{G(2)}, \lambda_{G(3)}\}$ and their eigenvectors. For Type I regions, $\lambda_{G(0)}$ corresponds to $-\kappa\varrho_p$, while $\lambda_{G(1)}, \lambda_{G(2)}, \lambda_{G(3)}$ correspond to $\kappa p_{(r)}, \kappa p_{(\theta)}, \kappa p_{(\phi)}$ respectively, based on the eigenvector projections detailed in the physical reordering. For Type IV regions, $\lambda_{G(0)}, \lambda_{G(1)}$ are the complex pair $\xi \pm i\mu$, and $\lambda_{G(2)}, \lambda_{G(3)}$ are the sorted real eigenvalues corresponding to the transverse pressures.

Reliability and precision

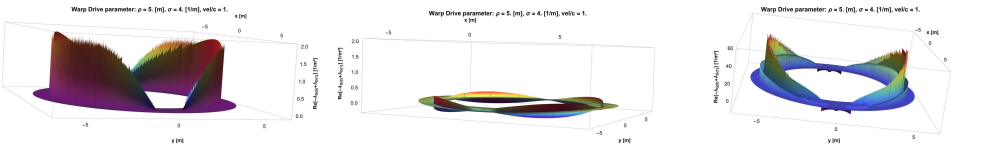
For every initially computed eigenpair $\{\lambda_I, \boldsymbol{v}_I\}_{I=1..4}$, we verify the residual $\|M\boldsymbol{v}_I - \lambda_I \boldsymbol{v}_I\|_2 = \mathscr{O}(10^{-p})$ at the working precision p. The numerical matrix $M = G^{\hat{\alpha}}{}_{\hat{\beta}}$, derived from real tetrad components, is itself real; any non-real eigenvalues occur in complex conjugate pairs. Reliability is further supported by verifying that the sum of the final, physically reordered eigenvalues, $\sum_{\alpha=0}^{3} \lambda_{G(\alpha)}$, equals the trace $\operatorname{Tr} M = G^{\hat{\beta}}{}_{\hat{\beta}} = G^{(\beta)}{}_{(\beta)} = G = \kappa T = -R$ (the scalar curvature invariant) and is purely real to $\mathscr{O}(10^{-p})$ accuracy. Exploratory parameter scans are performed at machine precision ($p \approx 16$ decimal digits), whereas all quoted numbers and the figures employ $p = 50$.



(a) Alcubierre: $-\operatorname{Re}[\lambda_{G(0)}]$ (b) Irrotational: $-\operatorname{Re}[\lambda_{G(0)}]$ (c) Natário: $-\operatorname{Re}[\lambda_{G(0)}]$

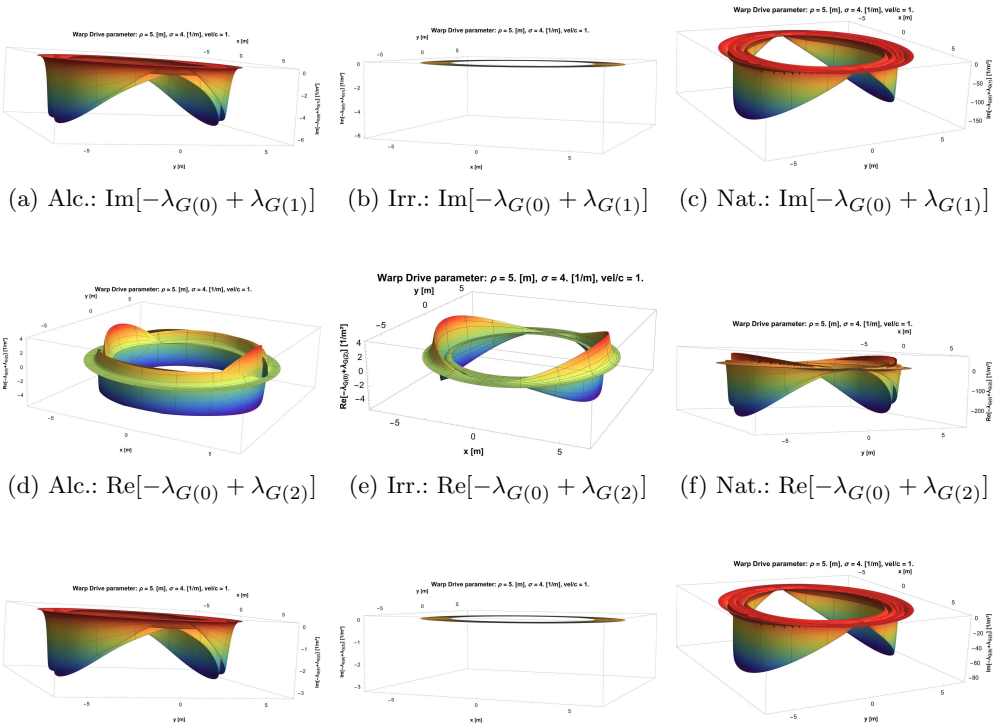


(d) Alcubierre: $-\operatorname{Im}[\lambda_{G(0)}]$ (e) Irrotational: $-\operatorname{Im}[\lambda_{G(0)}]$ (f) Natário: $-\operatorname{Im}[\lambda_{G(0)}]$



(g) Alc.: $\operatorname{Re}[-\lambda_{G(0)} + \lambda_{G(1)}]$ (h) Irr.: $\operatorname{Re}[-\lambda_{G(0)} + \lambda_{G(1)}]$ (i) Nat.: $\operatorname{Re}[-\lambda_{G(0)} + \lambda_{G(1)}]$

Fig. 2: Distribution of physically reordered eigenvalues $\lambda_{G(0)}$ and $\lambda_{G(1)}$ against the axis of travel x and transverse y (coordinates in Fig. 1). Columns compare Alcubierre, irrotational, and Natário warp drives. Rows display: (1) $-\operatorname{Re}[\lambda_{G(0)}]$; (2) $-\operatorname{Im}[\lambda_{G(0)}]$; (3) $\operatorname{Re}[-\lambda_{G(0)} + \lambda_{G(1)}]$. In Type I regions, $\operatorname{Im}[\lambda_{G(0)}] = 0$, hence $-\lambda_{G(0)}$ (in row (1)), equals κ times the *proper* energy density; conversely, in Type IV regions, $-\operatorname{Im}[\lambda_{G(0)}]$(in row (2)), is non-zero and proportional to the energy flux. See text for detailed physical interpretation and NEC analysis.



(a) Alc.: $\mathrm{Im}[-\lambda_{G(0)} + \lambda_{G(1)}]$ (b) Irr.: $\mathrm{Im}[-\lambda_{G(0)} + \lambda_{G(1)}]$ (c) Nat.: $\mathrm{Im}[-\lambda_{G(0)} + \lambda_{G(1)}]$

(d) Alc.: $\mathrm{Re}[-\lambda_{G(0)} + \lambda_{G(2)}]$ (e) Irr.: $\mathrm{Re}[-\lambda_{G(0)} + \lambda_{G(2)}]$ (f) Nat.: $\mathrm{Re}[-\lambda_{G(0)} + \lambda_{G(2)}]$

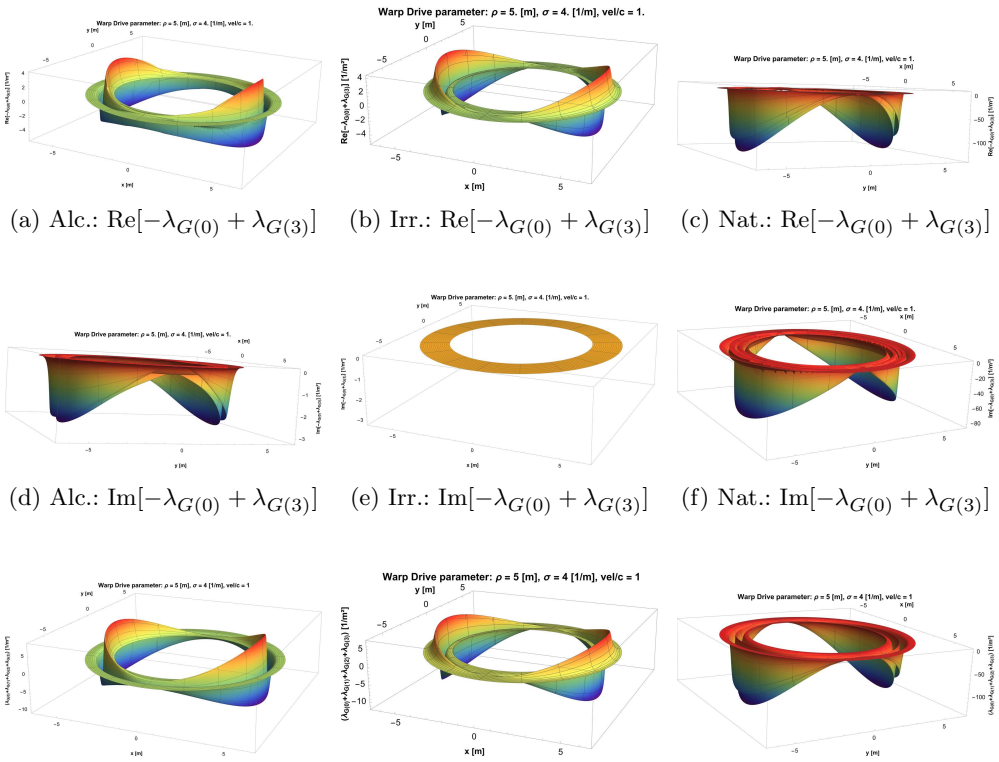
(g) Alc.: $\mathrm{Im}[-\lambda_{G(0)} + \lambda_{G(2)}]$ (h) Irr.: $\mathrm{Im}[-\lambda_{G(0)} + \lambda_{G(2)}]$ (i) Nat.: $\mathrm{Im}[-\lambda_{G(0)} + \lambda_{G(2)}]$

Fig. 3: Distribution of eigenvalue combinations relevant to energy condition analysis. Plotted against the axis of travel x and transverse coordinate y (coordinates in Fig. 1). Columns compare Alcubierre, irrotational, and Natário warp drives. Rows display: (1) $\mathrm{Im}[-\lambda_{G(0)} + \lambda_{G(1)}]$; (2) $\mathrm{Re}[-\lambda_{G(0)} + \lambda_{G(2)}]$; (3) $\mathrm{Im}[-\lambda_{G(0)} + \lambda_{G(2)}]$. See text for null energy condition (NEC) interpretations.

4.2 Interpretation of eigenvalues and energy conditions from visualizations

Three-dimensional *Mathematica*® [34] `RevolutionPlot3D` surfaces (Figs. 2–4) display the meridional distribution of selected combinations of the *physically sorted eigenvalues* $\{\lambda_{G(\alpha)}\}_{\alpha=0}^{3}$ that enter the weak (WEC) and null (NEC) energy-condition diagnostics, together with the scalar *invariant* $G = -R = \sum_{\alpha=0}^{3} \lambda_{G(\alpha)}$. Each surface sits in the prime meridional (x, y)-plane (the x-axis points along the direction of motion; cf. Fig. 1); with its vertical coordinate z the amplitude of the plotted quantity, encoded by a rainbow color scale keyed to the z-values.

Alcubierre and irrotational results are plotted on a common vertical scale. Natário requires its own scale because the zero-expansion shift field forces amplitudes larger



(a) Alc.: $\mathrm{Re}[-\lambda_{G(0)} + \lambda_{G(3)}]$ (b) Irr.: $\mathrm{Re}[-\lambda_{G(0)} + \lambda_{G(3)}]$ (c) Nat.: $\mathrm{Re}[-\lambda_{G(0)} + \lambda_{G(3)}]$

(d) Alc.: $\mathrm{Im}[-\lambda_{G(0)} + \lambda_{G(3)}]$ (e) Irr.: $\mathrm{Im}[-\lambda_{G(0)} + \lambda_{G(3)}]$ (f) Nat.: $\mathrm{Im}[-\lambda_{G(0)} + \lambda_{G(3)}]$

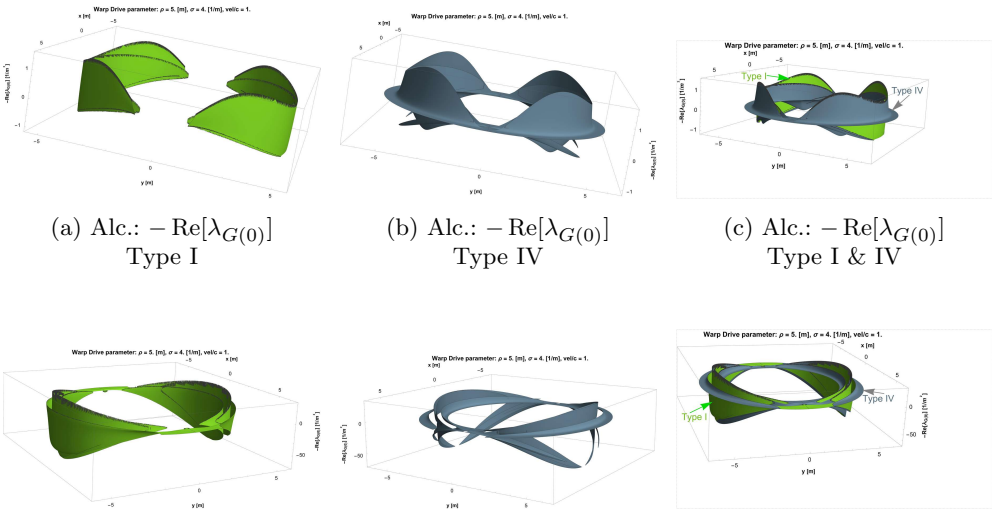
(g) Alc. $G = \sum_{\sigma=0}^{3} \lambda_{G(\sigma)}$ (h) Irr. $G = \sum_{\sigma=0}^{3} \lambda_{G(\sigma)}$ (i) Nat. $G = \sum_{\sigma=0}^{3} \lambda_{G(\sigma)}$

Fig. 4: Distribution of further eigenvalue combinations. Plotted against the axis of travel x and transverse coordinate y (coordinates in Fig. 1). Columns compare Alcubierre, irrotational, and Natário warp drives. Rows display: (1) $\mathrm{Re}[-\lambda_{G(0)} + \lambda_{G(3)}]$; (2) $\mathrm{Im}[-\lambda_{G(0)} + \lambda_{G(3)}]$; (3) Trace invariant $G = \sum_{\sigma=0}^{3} \lambda_{G(\sigma)}$. See text for physical interpretation and energy condition analysis.

(by an order of magnitude), which underscores how suppressing local spatial volume expansion forces much larger curvature—and hence much larger stress–energy densities—to sustain the warp bubble.

To visualize the global Hawking–Ellis type distribution of the Alcubierre and Natário bubbles, Fig. 5 —also generated with `RevolutionPlot3D`— plots the timelike-associated eigenvalue $-\mathrm{Re}[\lambda_{G(0)}]$ with color encoding algebraic type (light for Type I, dark for Type IV). Regions are classified as Type I ($\mathrm{Im}[\lambda_{G(0)}] = 0$, hence $-\mathrm{Re}[\lambda_{G(0)}] = \kappa\, \varrho_p$) versus Type IV ($\mathrm{Im}[\lambda_{G(0)}] \equiv \mu \neq 0$). Columns display (left) Type I only, (center) Type IV only, and (right) their overlay; rows distinguish (top) Alcubierre from (bottom) Natário.

Complementary one-dimensional radial cuts—generated with `Plot`—along the axis of motion ($\theta = 0$) and in the equatorial plane ($\theta = \pi/2$) (Figs. 6–9) plot each



(a) Alc.: $-\mathrm{Re}[\lambda_{G(0)}]$ Type I

(b) Alc.: $-\mathrm{Re}[\lambda_{G(0)}]$ Type IV

(c) Alc.: $-\mathrm{Re}[\lambda_{G(0)}]$ Type I & IV

(d) Nat.: $-\mathrm{Re}[\lambda_{G(0)}]$ Type I

(e) Nat.: $-\mathrm{Re}[\lambda_{G(0)}]$ Type IV

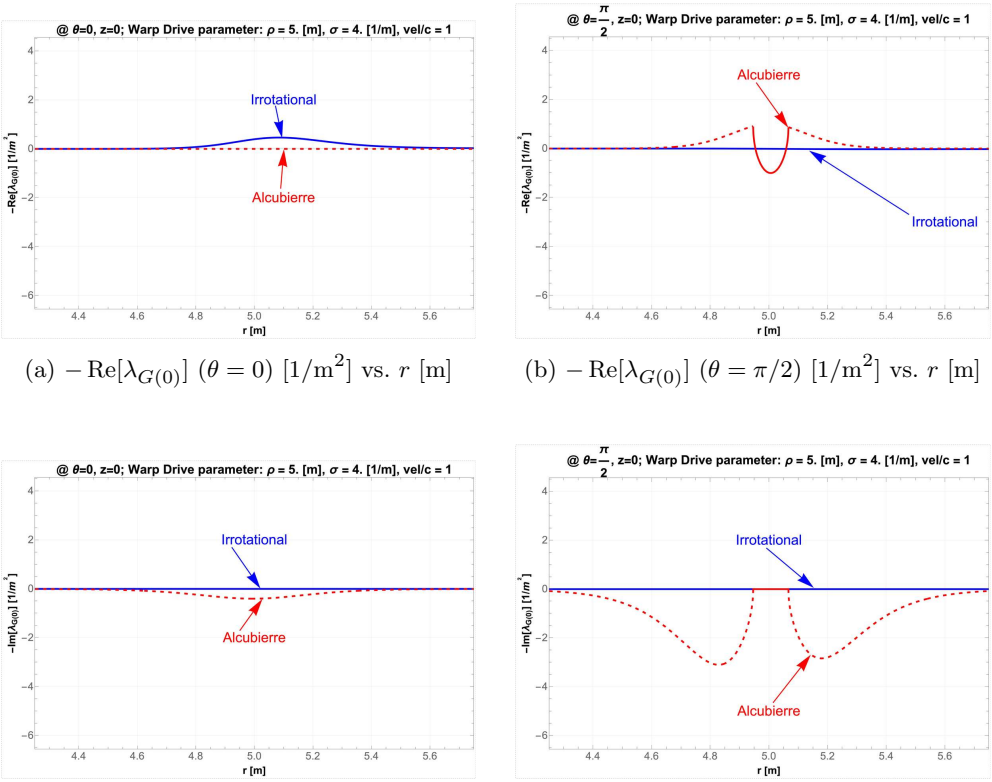
(f) Nat.: $-\mathrm{Re}[\lambda_{G(0)}]$ Type I & IV

Fig. 5: Hawking–Ellis classification of the timelike-associated eigenvalue $-\mathrm{Re}[\lambda_{G(0)}]$ (where $\varrho_p = -\mathrm{Re}[\lambda_{G(0)}]/\kappa$ in Type I regions), plotted in the prime meridional x–y plane (with x the direction of motion and y the transverse axis) (convention in Fig. 1). Columns depict regions classified as: (Left) Type I; (Center) Type IV; (Right) overlays both types. Rows correspond to: (1) Alcubierre's warp drive; (2) Natário's warp drive.

physically-sorted eigenvalue $\{\lambda_{G(\alpha)}\}_{\alpha=0}^{3}$ across the warp-bubble walls. Line style indicates Hawking–Ellis classification (solid for Type I, dashed for Type IV), for direct comparison with Fig. 5.

The 3-D type-structure surfaces and 1-D cuts collectively reveal the persistent Type IV regions in the Alcubierre and Natário drives versus the irrotational solution's global Type I character. As shown in theorem 2, the irrotational warp drive therefore admits a local rest frame, a unique timelike eigenvector, and an *invariant* proper energy density.

The first two eigenvalues in the physically-sorted list, $\lambda_{G(0)}$ and $\lambda_{G(1)}$, correspond, in the case of Type IV regions, to the complex conjugate pair $\Lambda_G = \xi \pm i\mu$. This implies that their Type IV regions, as discussed within section 4.1.2, possess no real causal eigenvectors, admit no preferred local rest frame, and lack a uniquely defined invariant proper energy density (cf. Table 2). These characteristics are consequences of the non-zero parameter μ, which corresponds to an irreducible energy flux component $q_s = \mu/\kappa$ within the invariant 2-plane associated with the complex eigenvalues. The null energy condition (NEC), expressed in terms of the mixed Einstein tensor tetrad components as $G^{\hat{\sigma}}{}_{\hat{\beta}}\eta_{\hat{\alpha}\hat{\sigma}}k^{\hat{\alpha}}k^{\hat{\beta}} \geq 0$ for all null vectors $k^{\hat{\alpha}}$ (satisfying $\eta_{\hat{\gamma}\hat{\delta}}k^{\hat{\gamma}}k^{\hat{\delta}} = 0$), is inherently violated in regions where $\mu \neq 0$. This violation occurs because the Type IV algebraic structure guarantees the condition fails for some null directions $k^{\hat{\alpha}}$. Since the



(a) $-\mathrm{Re}[\lambda_{G(0)}]$ $(\theta = 0)$ $[1/\mathrm{m}^2]$ vs. r [m]

(b) $-\mathrm{Re}[\lambda_{G(0)}]$ $(\theta = \pi/2)$ $[1/\mathrm{m}^2]$ vs. r [m]

(c) $-\mathrm{Im}[\lambda_{G(0)}]$ $(\theta = 0)$ $[1/\mathrm{m}^2]$ vs. r [m]

(d) $-\mathrm{Im}[\lambda_{G(0)}]$ $(\theta = \pi/2)$ $[1/\mathrm{m}^2]$ vs. r [m]

Fig. 6: Real and imaginary parts of the timelike-dominant eigenvalue $-\lambda_{G(0)}$ $(\theta = 0, \pi/2)$ vs. radial distance r for Alcubierre and irrotational warp drives. Line style indicates local Hawking–Ellis classification: **solid** for Type I, **dashed** for Type IV (null complex pair).

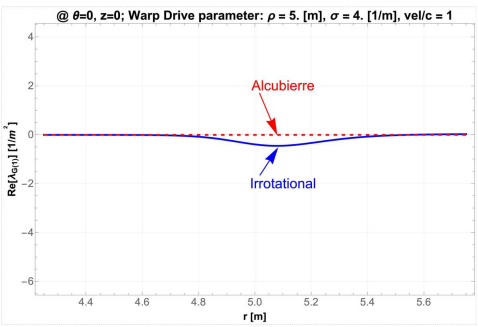
weak energy condition (WEC) implies the NEC, the WEC is also necessarily violated in these Type IV regions.

4.2.1 Timelike eigenvalue and energy density interpretation

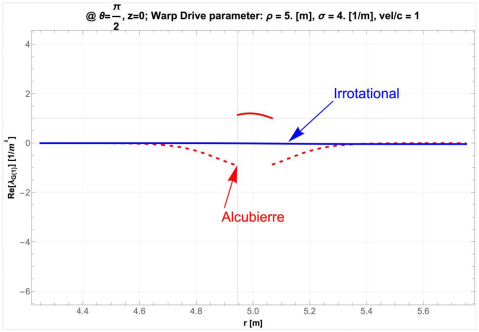
We now detail the interpretation of the eigenvalue $-\lambda_{G(0)}$ visualizations (Figs. 2–9), starting with the irrotational drive.

Irrotational warp drive

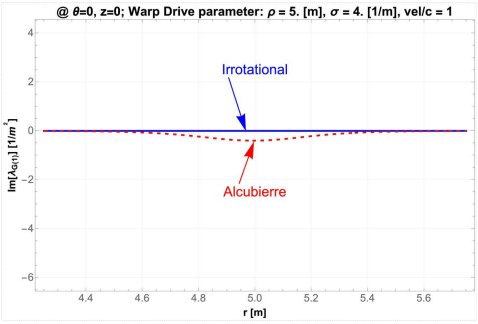
The irrotational warp drive's global Type I classification ensures its eigenvalues are real (e.g., $\mathrm{Im}[\lambda_{G(0)}] = 0$, Fig. 2e). The quantity $-\lambda_{G(0)}$, related to the invariant proper energy density by $-\lambda_{G(0)} = \kappa\varrho_p$ and depicted in Fig. 2b, must accordingly be non-negative ($-\lambda_{G(0)} \geq 0$) for the weak energy condition (WEC, requiring $\varrho_p \geq 0$) to be satisfied. Calculations show this quantity $-\lambda_{G(0)}$ is predominantly positive (reaching



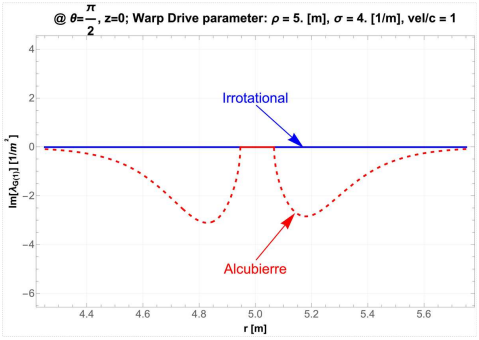
(a) $\mathrm{Re}[\lambda_{G(1)}]$ $(\theta = 0)$ $[1/\mathrm{m}^2]$ vs. r [m]



(b) $\mathrm{Re}[\lambda_{G(1)}]$ $(\theta = \pi/2)$ $[1/\mathrm{m}^2]$ vs. r [m]



(c) $\mathrm{Im}[\lambda_{G(1)}]$ $(\theta = 0)$ $[1/\mathrm{m}^2]$ vs. r [m]



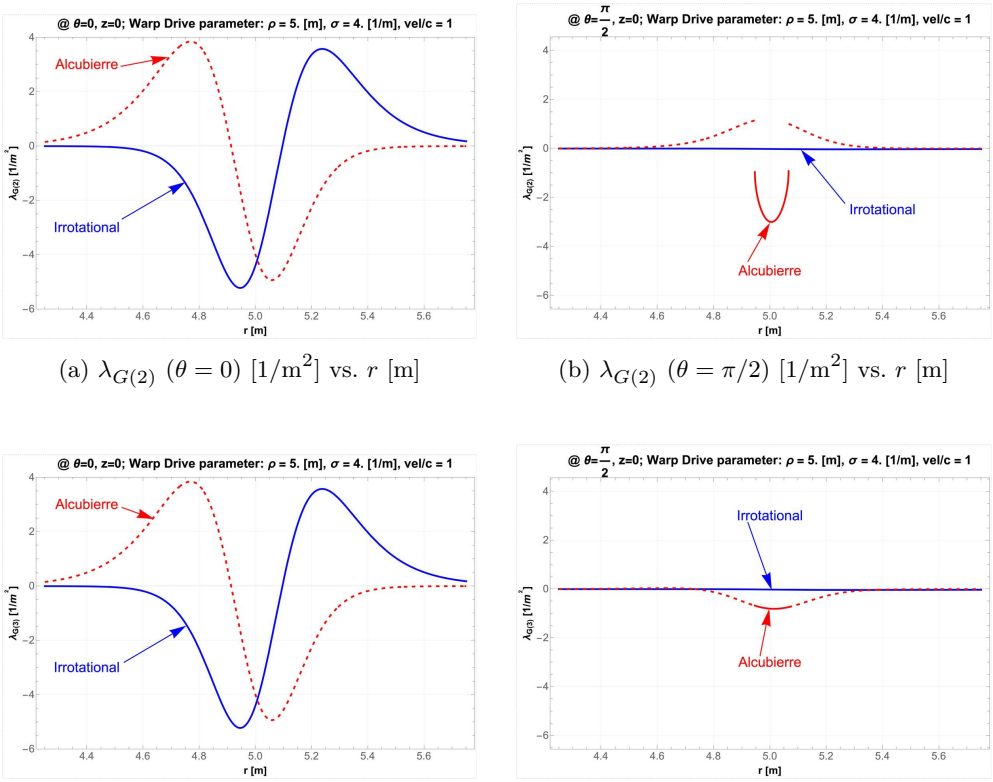
(d) $\mathrm{Im}[\lambda_{G(1)}]$ $(\theta = \pi/2)$ $[1/\mathrm{m}^2]$ vs. r [m]

Fig. 7: Real and imaginary parts of the spacelike-dominant eigenvalues in the travel direction, $\lambda_{G(1)}$ $(\theta = 0, \pi/2)$, vs. radial distance r for Alcubierre and irrotational warp drives. Line style indicates local Hawking–Ellis classification: **solid** for Type I, **dashed** for Type IV (null complex pair).

a maximum of $0.46\,[\mathrm{m}^{-2}]$ on the axis of motion, $\theta = 0$), but becomes slightly negative (reaching a minimum of $-0.026\,[\mathrm{m}^{-2}]$) in a localized region near $\theta = \pi/2$ (perpendicular to the direction of travel). This indicates a small, localized WEC violation ($\varrho_p < 0$) at $\theta = \pi/2$, undiscernible at the scale shown in the figure.

Alcubierre warp drive

Global algebraic features. Both the Alcubierre spacetime (Figs. 5b, 5c) and the Natário spacetime (Figs. 5e, 5f) contain extended Type IV regions, where the mixed Einstein tensor $G^{\hat{\mu}}{}_{\hat{\nu}}$ possesses complex eigenvalues $\Lambda_G = \xi \pm i\mu$ with $\mu \neq 0$ (Figs. 2d, 2f). This algebraic structure *necessarily violates* the null and weak energy conditions (see Table 2): no real timelike eigenvector exists, the observer-dependent energy density in the invariant two-plane is $\varrho_{\mathrm{obs}} = -\xi/\kappa < 0$, and the irreducible energy flux is $q_s = \mu/\kappa$.



(a) $\lambda_{G(2)}$ $(\theta = 0)$ $[1/\mathrm{m}^2]$ vs. r [m]

(b) $\lambda_{G(2)}$ $(\theta = \pi/2)$ $[1/\mathrm{m}^2]$ vs. r [m]

(c) $\lambda_{G(3)}$ $(\theta = 0)$ $[1/\mathrm{m}^2]$ vs. r [m]

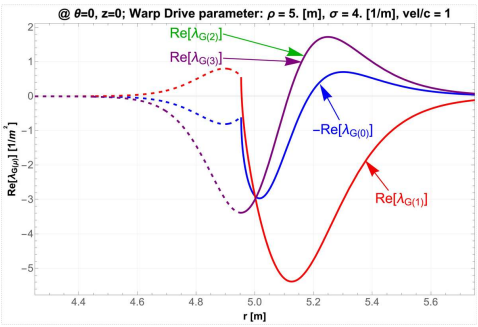
(d) $\lambda_{G(3)}$ $(\theta = \pi/2)$ $[1/\mathrm{m}^2]$ vs. r [m]

Fig. 8: Spacelike-dominant purely real eigenvalues perpendicular to the travel direction: $\lambda_{G(2)}$ $(\theta = 0, \pi/2)$ and $\lambda_{G(3)}$ $(\theta = 0, \pi/2)$ vs. radial distance r for Alcubierre and irrotational warp drives. Line style indicates local Hawking–Ellis classification: **solid** for Type I, **dashed** for Type IV (null complex pair).

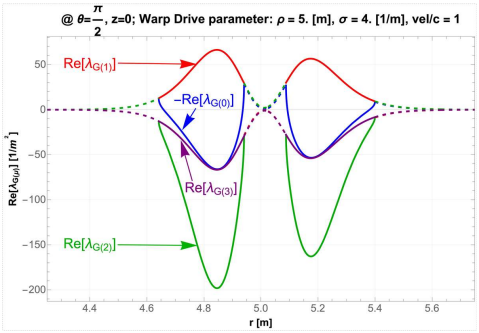
Axis of motion cut ($\theta = 0$). Along the axis of motion the tensor is Type IV for all radii (Figs. 6a, 6c). The imaginary part dominates, $|\mu| > |\xi|$, but its absolute value is *not* maximal on this cut (Fig. 6c).

Equatorial plane cut ($\theta = \pi/2$). Along this cut the algebraic type varies with radius. In the bubble interior ($r \lesssim \rho - \Delta$) and the far field ($r \gg \rho + \Delta$), where $\Delta \approx 0.014\rho$ the tensor remains Type IV; the energy flux parameter $q_s = \mu/\kappa$ attains its *largest* magnitude, $\mu \simeq -3.0$ $[\mathrm{m}^{-2}]$ at $r \approx 0.96\rho$ (Fig. 6d). Within an annulus of width $\sim 2\Delta$ about the wall ($r \approx \rho$) we have ($\mu = 0$) and the tensor switches to Type I. A unique rest frame exists, and the timelike eigenvalue $\kappa\, \varrho_p(r) = -\lambda_{G(0)}(r)$ acquires a pronounced U-shape: it peaks at $\kappa\, \varrho_p \simeq 0.9$ $[\mathrm{m}^{-2}]$ near $r \approx \rho \pm \Delta$ and dips to $\kappa\, \varrho_p \simeq -1.0$ $[\mathrm{m}^{-2}]$ at $r \approx \rho$ (Fig. 6b), producing a local WEC violation.

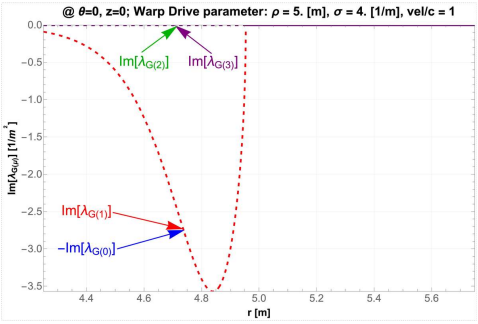
Comparative energy-density profiles of Alcubierre and irrotational drives. The irrotational solution is *everywhere* Type I, with its peak positive proper energy density



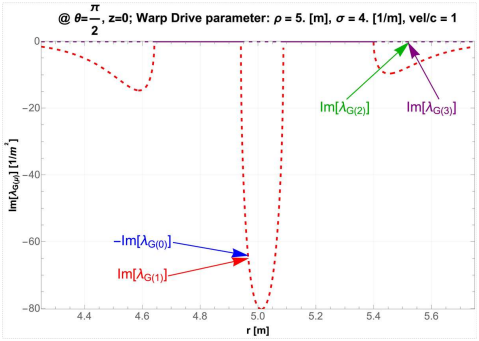
(a) Real parts $\mathrm{Re}(\lambda_{G(\mu)})$ vs. r at $\theta = 0$.



(b) Real parts $\mathrm{Re}(\lambda_{G(\mu)})$ vs. r at $\theta = \pi/2$.



(c) Imaginary parts $\mathrm{Im}(\lambda_{G(\mu)})$ vs. r at $\theta = 0$.



(d) Imaginary parts $\mathrm{Im}(\lambda_{G(\mu)})$ vs. r at $\theta = \pi/2$.

Fig. 9: Physically sorted eigenvalues $\{\lambda_{G(\mu)}\}_{\mu=0}^{3}$ of $G^{\hat{\alpha}}{}_{\hat{\beta}}$ versus radial distance r for Natário's warp drive. Panels display real parts (top row) and imaginary parts (bottom row) of the eigenvalues along the axis of motion ($\theta = 0$, left column) and in the equatorial plane ($\theta = \pi/2$, right column). Line style indicates local Hawking–Ellis classification: **solid** for Type I (all eigenvalues real), **dashed** for Type IV (complex eigenvalues present).

$\varrho_p \simeq 0.46\,[\mathrm{m}^{-2}]/\kappa$ on the axis of motion ($\theta = 0,\ r \approx \rho$). By contrast, the Alcubierre drive's Type I region with peak negative proper density $\varrho_p \simeq -1.0\,[\mathrm{m}^{-2}]/\kappa$ occurs at the equatorial cut (at $\theta = \pi/2,\ r \approx \rho$). Therefore the irrotational peak positive proper energy density is 46 % of that Alcubierre magnitude. Furthermore, the irrotational peak ϱ_p is only about 15% of the Alcubierre drive's peak irreducible flux magnitude $|q_s| \simeq 3.0\,[\mathrm{m}^{-2}]/\kappa$ in its equatorial Type IV region; this flux magnitude itself is roughly 3 times larger than the magnitude of the Alcubierre drive's peak negative ϱ_p in its Type I zone. Comparing the proper energy density $|\varrho_p|$ of the Type I irrotational drive with the irreducible flux magnitude $|q_s|$ from Alcubierre's Type IV regions provides a useful gauge of relative *exotic matter cost*, despite their distinct physical roles. Both quantities are scalar invariants with the same physical dimensions ($[\mathrm{m}^{-2}]$). As

established in theorem 3, for Type IV regions, $|\mu|$ (and thus $|q_s|$) is invariant, while $\xi = \mathrm{Re}[\lambda_{G(0)}]$ is an invariant scalar that relates to an observer-dependent energy density $\varrho_{\mathrm{obs}} = -\xi/\kappa$ for observers within the null two-plane $\mathscr{N}$; such regions inherently lack a unique rest frame and proper energy density. In contrast, the irrotational drive's Type I structure yields a true frame-invariant proper energy density $\varrho_p = -\lambda_{G(0)}/\kappa$.

Thus, eliminating vorticity ($\omega_{ij} = 0$) while allowing non-zero expansion leads to a dramatically different energy-density profile for the irrotational drive. The requirements are substantially smaller in absolute magnitude (e.g., its peak ϱ_p is $\approx 15\%$ of Alcubierre's peak $|q_s|$, a reduction factor of nearly 7), and the required stress-energy is predominantly positive and algebraically Type I (classical energy-density) everywhere, unlike the large negative densities and Type IV (complex-null) zones characteristic of the Alcubierre drive.

Natário warp drive

Figure 9 displays the eigenvalue behavior for the Natário warp drive along radial cuts at $\theta = 0$ (axis of motion) and $\theta = \pi/2$ (equatorial plane). The Natário drive exhibits radially varying Hawking–Ellis classifications along both cuts shown.

Along the axis ($\theta = 0$) (Figs. 9a and Fig. 9c), the geometry transitions from Type IV for radii approximately $r \lesssim \rho$ (indicated by a non-zero imaginary component $\mu = \mathrm{Im}[\lambda_{G(0)}]$) to Type I for $r \gtrsim \rho$ (where $\mu = 0$ and eigenvalues become purely real).

The equatorial cut ($\theta = \pi/2$), reveals a more complex sequence of alternating classifications:

- **Type IV**: ($\mu \neq 0$), from the origin $r = 0$ to the inner wall,
- **Type I**: ($\mu = 0$), inner wall, with the proper energy density peaking at $\varrho_p = -\xi/\kappa \simeq -67\,[\mathrm{m}^{-2}]/\kappa$(Fig. 9b), i.e. 146 times the irrotational drive's peak magnitude.
- **Type IV**: ($\mu \neq 0$) the central region extending over $r \approx \rho$, where $\mu = \mathrm{Im}[\lambda_{G(0)}] \simeq -80\,[\mathrm{m}^{-2}]$ (Fig. 9d), i.e. 174 times the irrotational drive's peak.
- **Type I**: ($\mu = 0$) outer wall.
- **Type IV**: ($\mu \neq 0$) in the far field ($r \to \infty$).

The peak eigenvalue magnitudes (both the real part ξ and imaginary part μ) in the equatorial cut ($\theta = \pi/2$) exceed those along the axis of travel ($\theta = 0$), and are significantly larger than those required for the Alcubierre and irrotational drives, reflecting the comparatively higher exotic-matter cost of the Natário solution, due to its zero volume expansion constraint.

4.2.2 Null energy condition interpretation

We now detail the interpretation of the null energy condition (NEC) based on the eigenvalue visualizations (Figs. 2–4). The NEC requires that $T_{\mu\nu}k^\mu k^\nu \geq 0$, or equivalently using the mixed Einstein tensor tetrad components central to our analysis, $G^{\hat{\sigma}}{}_{\hat{\beta}} k_{\hat{\sigma}} k^{\hat{\beta}} \geq 0$ for all null vectors $k^{\hat{\beta}}$ (where $k_{\hat{\sigma}} = \eta_{\hat{\sigma}\hat{\alpha}} k^{\hat{\alpha}}$ and the null condition is $\eta_{\hat{\gamma}\hat{\delta}} k^{\hat{\gamma}} k^{\hat{\delta}} = 0$). For Type I regions, the NEC implies $\varrho_p + p_k \geq 0$ for all principal pressures p_k. Given the relations $\kappa\,\varrho_p = -\lambda_{G(0)}$ and $\kappa\,p_k = \lambda_{G(k)}$ adopted here (for $k = 1, 2, 3$), the NEC condition for Type I regions becomes $\lambda_{G(k)} - \lambda_{G(0)} \geq 0$.

The plots focusing on relevant eigenvalue combinations (Figs. 3d-3f, 4a-4c) display the quantities $\lambda_{G(k)} - \mathrm{Re}[\lambda_{G(0)}]$ (where $\xi = \mathrm{Re}[\lambda_{G(0)}]$ and $k = 2, 3$). For all three drives, the most significant NEC violations correspond to regions where these quantities become most negative. This typically occurs where the real eigenvalues $\lambda_{G(2)}, \lambda_{G(3)}$, associated with principal pressures (transverse based on physical sorting), reach their most negative values relative to ξ.

Irrotational warp drive

Applying this to the globally Type I irrotational drive, Figs. 3 and 4 indicate that the most significant NEC violations (the most negative values of $\lambda_{G(k)} - \lambda_{G(0)}$), approximately $-4.[\mathrm{m}^{-2}]$, occur along the axis of motion ($\theta = 0$) associated with $\lambda_{G(2)} - \lambda_{G(0)}$ (Fig. 3e) and with $\lambda_{G(3)} - \lambda_{G(0)}$ (Fig. 4b). The origin of this violation is clarified by Figs. 8a and 8c, which show that the transverse pressure eigenvalues $\lambda_{G(2)}$ and $\lambda_{G(3)}$ themselves become significantly negative along this axis. These large transverse negative pressures ($p_k = \lambda_{G(k)}/\kappa < 0$) are responsible for $\varrho_p + p_k < 0$, thus violating the NEC, even though ϱ_p is positive at $\theta = 0$ (consistent with $-\lambda_{G(0)}$ peaking positively there).

Alcubierre warp drive

In the Type IV regions characteristic of the Alcubierre and Natário drives, null energy condition (NEC) violation is inherently guaranteed by the algebraic structure associated with $\mu \neq 0$ (cf. section 4.1). The observation from Figs. 3d-3f and 4a-4c that the quantities $\lambda_{G(k)} - \xi$ (where $\xi = \mathrm{Re}[\lambda_{G(0)}]$ and $k = 2, 3$) become strongly negative indicates directions of particularly strong NEC violation. This arises from the interplay between terms related to the real eigenvalues $\lambda_{G(k)}$ (associated with $k = 2, 3$ transverse negative principal pressures) and ξ.

For the Alcubierre drive, specifically, Fig. 3d indicates that significant NEC violations occur practically uniformly in the xy prime meridian plane, both in the cut along the axis of motion ($\theta = 0$) and in the equatorial cut ($\theta = \pi/2$). At the $\theta = 0$ location, the geometry is Type IV, as confirmed by the eigenvalue behavior shown in the radial cuts in the direction of motion, Figs. 8a and 8c. Here, the peak NEC violation of approximately $-4.3[\mathrm{m}^{-2}]$ is driven by the transverse pressure-related eigenvalues $\lambda_{G(2)}$ and $\lambda_{G(3)}$ becoming significantly negative relative to ξ. The Type IV classification of the Alcubierre drive at this location contrasts with the irrotational and Natário drives, which are classified as Type I at the regions of largest NEC violation.

Natário warp drive

Focusing on the Natário drive's equatorial cut ($\theta = \pi/2$), Fig. 3f reveals a maximum null energy condition (NEC) violation, where the relevant eigenvalue combination reaches a minimum $\lambda_{G(2)} - \lambda_{G(0)} \approx -260\,[\mathrm{m}^{-2}]$. This large negative value signifies a strong breach of the physical NEC requirement $\varrho_p + p_k \geq 0$, specifically $\varrho_p + p_2 \ll 0$ here. Interestingly, at this radial location of maximum violation, the geometry is locally Type I ($\mu = 0$, Fig. 9d). The violation $\varrho_p + p_2 \ll 0$ occurs because a strongly negative principal pressure ($p_2 = \lambda_{G(2)}/\kappa \approx -200[\mathrm{m}^{-2}]/\kappa$) (Fig. 9b) adds to the proper energy density, which is also negative here ($\varrho_p = -\lambda_{G(0)}/\kappa \approx -60[\mathrm{m}^{-2}]/\kappa$, corresponding to

$\lambda_{G(0)} \approx +60\,[\mathrm{m}^{-2}]$) (Fig. 9b). The peak NEC violation magnitude of $\approx -260\,[\mathrm{m}^{-2}]$ for the Natário drive significantly exceeds (e.g., by a factor exceeding 60 times) the maximum violations found for the irrotational and Alcubierre drives.

4.2.3 Weak, dominant, and strong energy conditions

For all three warp–drive geometries we found a violation of the null energy condition (NEC). Since

$$\mathrm{DEC} \;\Rightarrow\; \mathrm{WEC} \;\Rightarrow\; \mathrm{NEC}, \qquad \mathrm{SEC} \;\Rightarrow\; \mathrm{NEC}, \tag{56}$$

their contrapositives give

$$\neg\mathrm{NEC} \;\Longrightarrow\; \neg\mathrm{WEC} \;\wedge\; \neg\mathrm{DEC} \;\wedge\; \neg\mathrm{SEC}. \tag{57}$$

Therefore all three warp–drive spacetimes simultaneously violate the weak (WEC), dominant (DEC), and strong (SEC) energy conditions.

4.2.4 Trace energy condition interpretation

The trace energy condition (TEC) [42, 43] is imposed on the trace of the stress-energy tensor. This invariant scalar can be expressed as $T \equiv T^{\hat{\alpha}}{}_{\hat{\alpha}}$. For our $(-,\,+,\,+,\,+)$ metric signature, the TEC requires this trace to be non-positive, i.e., $T \leq 0$. This is equivalent to the physical requirement that the proper energy density ϱ_p must be greater than or equal to the sum of the principal pressures p_k, $\varrho_p \geq \sum_k p_k$. Through Einstein's field equations, $G^{\hat{\alpha}}{}_{\hat{\beta}} = \kappa\, T^{\hat{\alpha}}{}_{\hat{\beta}}$, this condition on T directly translates to $G \equiv G^{\hat{\alpha}}{}_{\hat{\alpha}}$, via $G = \kappa\, T$. Consequently, the TEC ($T \leq 0$) implies a non-positive Einstein tensor scalar ($G \leq 0$) and thus a non-negative Ricci scalar ($R = -G \geq 0$). The scalar invariant $G = \sum_{\alpha=0}^{3} \lambda_{G(\alpha)}$ (equal to $\mathrm{Tr}\, M$) provided a valuable check on our numerical eigensystem calculations to confirm the precise cancellation of imaginary parts from any complex conjugate eigenvalues. Historically, the notion that matter might be constrained in a way consistent with the TEC was foreshadowed by proposals regarding its ultimate equation of state. J. von Neumann suggested (as noted in 1935 [44]) a limiting equation of state $p = (1/3)\varrho$, which precisely saturates the TEC requirement that $p \leq \rho/3$. The TEC was largely abandoned after 1961 when Zel'dovich [45] showed that plausible ultra-dense matter, characterized by stiff equations of state such as $p \approx \varrho_p$, can violate the TEC. Bekenstein [46] introduced in 2013 a subdominant trace energy condition (STEC) for ordinary (baryonic) matter, requiring its stress-energy to satisfy a TEC-like constraint ($\rho \gtrsim \sum_k p_k$) that dominates exotic contributions from quantum vacuum energy. Consequently, the STEC ensures that the positive mass-energy of any physical apparatus would outweigh localized negative vacuum energy it contains, precluding net negative total mass and demonstrating a role for modified TEC conditions in arguments for global mass positivity in semi-classical gravity. We visualize the behavior of the Einstein tensor scalar invariant G (Figs. 4g–4i) in this TEC context and now proceed to discuss its implications to further distinguish the drives.

Alcubierre and irrotational warp drives

Figures 4g and 4h indicate that the Alcubierre and irrotational drives exhibit similar spatial distributions and peak magnitudes for G. Both drives violate the TEC, with G attaining positive values exceeding $\approx 5\,[\mathrm{m}^{-2}]$, these maximum violations occurring along the axis of motion ($\theta = 0$).

Natário warp drive

Our analytic result (theorem 1) proves that the zero-expansion Natário solution satisfies the trace–energy condition everywhere: with the lapse fixed to $\alpha = 1$, a flat spatial metric ($R^{(3)} = 0$), and vanishing expansion $K = 0$, the Gauss–Codazzi scalar identity reduces to $R^{(4)} = K_{ij}K^{ij} \geq 0$ and hence $G = -R^{(4)} \leq 0$ point-wise.

The numerical profile (Fig. 4i) agrees, G ranging from 0 at the bubble centre to about $-120\,[\mathrm{m}^{-2}]$; the most negative values (strongest TEC satisfaction) occur perpendicular to the direction of motion ($\theta = \pi/2$). The non-positivity of $G = \sum_{\sigma=0}^{3} \lambda_{G(\sigma)}$ is governed by $\lambda_{G(2)}$: as Fig. 9b shows, this eigenvalue becomes strongly negative, giving a substantial negative pressure $p_{(2)} = \lambda_{G(2)}/\kappa$ and thereby enforcing $G \leq 0$ throughout the spacetime.

5 Limitations & Outlook

Scope assumptions. We work with unit lapse $\alpha = 1$, a flat and time–independent spatial metric, and an irrotational shift $\beta_i = \partial_i \Phi$ with constant v in the maps and global measures. These idealizations yield a clean Type I eigenstructure and enable direct model-to-model comparisons. Convergence as $r \to 0$ is analyzed separately in Appendix B.

Global budgets are mask–defined. All integrated energies and volumes use the disjoint regions $\mathscr{R}_-, \mathscr{R}_0, \mathscr{R}_+$ built from the Hessian scalar $\lambda_H = \frac{1}{2}(\mathrm{tr}\, H^2 - (\mathrm{tr}\, H)^2)$ with adaptive absolute thresholds (Appendix C). With our sign convention $T^{(0)}{}_{(0)} = -\varrho_p$ and $\kappa\, \varrho_p = -\lambda_H$, the reported magnitudes $E_\pm = \int_{\mathscr{R}_\pm} |\varrho_p|\, dV = \kappa^{-1} \int_{\mathscr{R}_\pm} |\lambda_H|\, dV$ and volumes $V_\pm, V_0$ are numerically robust; the zero band V_0 quantifies the tolerance deadband rather than physics. We integrate over an extended window $r \in [0, 12\rho]$ and report both finite–window ($R_{\mathrm{integ}} = 12\rho$) totals and *tail–corrected* ($R \to \infty$) estimates obtained via a two–point $1/R$ extrapolation using $R \in \{8\rho, 12\rho\}$ (App. C, Eq. (C.13)). Jacobian and mask–partition checks pass at roundoff.

Interpretation of "exotic matter". In the canonical run ($\rho = 5$ [m], $\sigma = 4$ [m^{-1}], $v/c = 1$) on the $R_{\mathrm{integ}} = 12\rho$ window we find $E_+/E_- = 1.07$ (3 s.f.) and, on that finite window, a negative–energy volume fraction near 70% (Tables C2–C3). After applying the $1/R$ tail extrapolation (App. C, Table C1), the *tail–corrected* magnitudes satisfy

$$E_-(\infty) = 1.33 \times 10^{44}\ \mathrm{J}, \qquad E_+(\infty) = 1.33 \times 10^{44}\ \mathrm{J},$$

so that the normalized net is $|E_{\mathrm{net}}|/E_{\mathrm{abs}}(\infty) = 0.04\%$ (four decimals in fractional units). Within our stated reporting precision (central values to 3 s.f., percentages to two decimals), the net proper energy is therefore *consistent with zero*; accordingly, we adopt $E_{\mathrm{net}}(\infty) = 0$ as the best estimate. Equatorial localization metrics show that

the negative part is diffuse and concentrated near $\theta \simeq \pi/2$, whereas the positive part forms high–amplitude, crescent–like thin collars near $\theta \simeq 0, \pi$.

What changes if assumptions relax. Letting α or v vary in time introduces explicit time dependence in K_{ij} and can modify averages and alignments, potentially reintroducing off–diagonal fluxes in mixed components and creating Type IV pockets. Departing from irrotational flow (adding vorticity) would break the simple Hessian reduction and alter λ_H–based diagnostics.

Outlook. (i) Extend to gently time–dependent $v(t)$ and $\alpha(t, \mathbf{x})$; (ii) couple to matter models consistent with Type I eigenstructure; (iii) assess averaged and quantum energy–condition constraints in semiclassical regimes.

6 Conclusions

This work has provided a detailed analytical and numerical study of a novel *kinematically irrotational* warp-drive spacetime within General Relativity. Building on the work of Santiago, Schuster, and Visser [3]—who showed that kinematically irrotational warp drives with unit lapse ($\alpha = 1$) and flat spatial slices have vanishing momentum density and therefore fall into the Hawking–Ellis Type I class—this paper advances the field by explicitly constructing, analyzing, and cross-comparing concrete warp-drive spacetimes with significantly improved properties.

The key new contributions presented in this paper, ordered by importance and novelty, are as follows:

1. We present an explicit, closed-form analytical construction of a kinematically irrotational warp drive—specified by the scalar potential $\Phi(r, \theta, t) = v(t)\, r\, g(r) \cos\theta$ (43) and the associated *continuous* shift components β^i (36), (42)—satisfying $\boldsymbol{\omega} = *(d\boldsymbol{\beta}^\flat) = 0$ with *smooth* boundary behavior, thereby avoiding the discontinuities of some prior proposals; to our knowledge, no previous work has provided a continuous, analytically derived irrotational shift of this form.
2. Causal evidence from a fixed–smoothing ablation. Holding the baseline smoothing profile $g(r)$ and mask tolerances fixed, we introduce a tunable vortical component and recompute the global budgets (Appendix D, Table D6). Even modest vorticity collapses the near-cancellation of proper energy: on the $R_{\text{integ}} = 12\rho$ window the ratio E_+/E_- falls below 10^{-2} by $\eta = 0.25$, and at $\eta = 1$ the negative-energy magnitude E_- increases by a factor > 500. This confirms that the favorable energy profile of our irrotational model is *caused* by its curl-free kinematics rather than by profile smoothing.
3. We develop a systematic, high-precision (50-digit) Cartan–tetrad numerical pipeline for pointwise eigenvalue–eigenvector analysis of the full Einstein tensor, incorporating Hawking–Ellis classification and energy-condition diagnostics. Running it on our closed-form irrotational solution provides an explicit, full-spacetime confirmation of a warp-drive geometry that is globally Type I—corroborating the demonstration of [3] and extending the zero-vorticity study of [4]. As an internal validation, the timelike eigenvalue obtained by direct diagonalization, $\lambda_{G(0)}$, agrees to roundoff with the independent Hessian-invariant evaluation λ_H (Appendix C; Eqs. (C.1), (C.3)).

4. We formally demonstrate (theorem 3) that for Hawking–Ellis Type IV (complex null) regions the real part ξ and the magnitude $|\mu|$ of the complex eigenvalues are Lorentz scalar invariants, thereby justifying invariant-based comparisons of the irreducible flux $|q_s| = |\mu|/\kappa$ and observer-dependent density $|\varrho_{\rm obs}| = |\xi|/\kappa$ with the Type I proper energy density ϱ_p.
5. We extend this numerical Hawking–Ellis classification analysis to the Alcubierre and Natário (zero-expansion) warp drives. While these spacetimes can exhibit localized Type IV regions, in our analysis none of the models investigated—including the newly constructed irrotational drive—contain any regions of Type II or Type III stress-energy classification.
6. We demonstrate that the irrotational warp drive achieves a predominantly *positive* proper energy density ϱ_p (peaking at $0.46\,[\mathrm{m}^{-2}]/\kappa$ on its axis of motion, $\theta = 0$), with only a small, localized dip below zero ($\varrho_p \approx -0.026\,[\mathrm{m}^{-2}]/\kappa$) at the equatorial cut ($\theta = \pi/2$). Relative to the *peak proper-energy deficit*, this minimum is reduced by a factor of ≈ 38 compared with Alcubierre ($\min \varrho_p \approx -1/\kappa$) and by $\approx 2.6 \times 10^3$ compared with Natário ($\min \varrho_p \approx -67/\kappa$). While these relative reductions are significant, the absolute stress–energy scales, $T^{\hat{\mu}}{}_{\hat{\nu}} = G^{\hat{\mu}}{}_{\hat{\nu}}/\kappa$, remain astronomically large within standard General Relativity. Crucially, the irrotational drive requires classical Type I energy rather than the complex-null Type IV exotic energy of the other models.
7. Whole–spacetime energy/volume summary. On the baseline window $R_{\rm integ} = 12\rho$ we find $E_+/E_- = 1.07$ (3 s.f.) and $V_-/(V_- + V_+) \approx 0.732$. After applying a two-point $1/R$ tail extrapolation with $R \in \{8\rho, 12\rho\}$ (App. C, Eq. (C.13), Table C1), the tail-corrected magnitudes are $E_-(\infty) = 1.33 \times 10^{44}$ J and $E_+(\infty) = 1.33 \times 10^{44}$ J, with $E_{\rm net}(\infty) = -1.12 \times 10^{41}$ J and $|E_{\rm net}|/E_{\rm abs}(\infty) = 0.04\%$ (four decimals in fractional units). Within our stated reporting precision, the net proper energy is *consistent with zero*; our best estimate is $E_{\rm net}(\infty) \approx 0$. As visualized in Appendix C, Fig. C1 (prime meridional slice), the positive energy density support is compact and concentrated near the bubble wall ($r \simeq \rho$), whereas the negative energy density is diffuse and significantly lower in amplitude.
8. All three warp–drive spacetimes violate the NEC—and hence the WEC, DEC, and SEC—due to large *negative transverse principal pressures*. Although superficially akin to the isotropic negative pressure of dark energy, these pressures are strongly anisotropic (and confined to the bubble wall). Quantitatively, the Natário drive's peak NEC violation is 65× and 60× larger than those in the irrotational and Alcubierre drives, respectively (nearly two orders of magnitude). Moreover, the location of maximal violation differs: for the irrotational drive it lies along the axis of motion ($\theta = 0$), whereas for both the Alcubierre and Natário drives it lies on the equatorial cut ($\theta = \pi/2$).
9. We present a comparative analysis of the trace energy condition (TEC), proving (theorem 1) that, under the stated assumptions, the Natário zero-expansion warp drive satisfies $G \leq 0$ throughout the domain considered, while both the Alcubierre and the new irrotational drives exhibit regions with $G > 0$.
10. We emphasize that, although one can mathematically impose both zero divergence ($\nabla \cdot \boldsymbol{\beta} = 0$) and zero vorticity ($\boldsymbol{\omega} = \nabla \times \boldsymbol{\beta} = 0$)—which forces $\boldsymbol{\beta} = -\nabla\Phi$

with Φ harmonic ($\nabla^2\Phi = 0$)—this severely limits the potentials available for a localized warp bubble. In contrast, our irrotational drive employs a deliberately non-harmonic Φ (so $\nabla\cdot\boldsymbol{\beta} = -\nabla^2\Phi \neq 0$), thereby opening a broader solution space and yielding markedly improved energy-condition profiles.

These results collectively advance the theoretical and numerical understanding of warp-drive spacetimes by delivering what is, to our knowledge, the first explicit, closed-form irrotational solution in an orthonormal-tetrad (Cartan) framework. Built on a smooth C^∞ form-function $f(r)$, our model yields globally continuous frame fields and spin coefficients, and exhibits markedly improved energy-condition profiles compared to both Alcubierre and Natário drives. Moreover, we provide the first comprehensive, pointwise Hawking–Ellis classification across all three geometries, and explain the origin of the discrete jumps in classification seen in Alcubierre and Natário spacetimes: although the Einstein tensor and its eigenvalues vary smoothly, the algebraic type flips abruptly whenever key thresholds are crossed (for example, when the imaginary part μ of a complex eigenvalue vanishes or when an eigenvector's causal character changes). In particular, the Type IV subtype—requiring strictly null complex eigenvectors—is structurally unstable in the usual sense: arbitrarily small perturbations generically drive μ away from zero. By combining these conceptual insights with our high-precision analytic–numerical Cartan-tetrad pipeline, we lay a robust foundation for future explorations of exotic spacetime geometries and the theoretical limits of faster-than-light travel in General Relativity. All quantitative statements above hold under the stated assumptions of unit lapse ($\alpha = 1$) and stationary flat spatial slices; relaxing these (e.g., during acceleration) can reintroduce momentum density and localized Type IV regions.

Declarations

- Funding No funding was received for conducting this study.
- Conflict of interest/Competing interests The author declares no competing interests.
- Ethics approval and consent to participate Not applicable
- Consent for publication Not applicable
- Data availability This is a theoretical study. All results are derived from the equations presented within the manuscript. Numerical data used to generate the figures are available from the author upon reasonable request.
- Materials availability Not applicable
- Code availability The custom Wolfram Mathematica scripts used for numerical calculations and figure generation are available from the author upon reasonable request.
- Author contribution The sole author was responsible for all aspects of this work, including conceptualization, methodology, analysis, and manuscript preparation.

Appendix A Notation, units, and coordinate system

This section introduces the notation and units used throughout this paper and presents the coordinate system.

- Spatial 3-tensors are denoted by the index (3) on the top left of the tensor: ${}^{(3)}R_{ij}$.
- Indices: Greek $\mu, \nu = 0 \ldots 3$ (spacetime), Latin $i, j = 1 \ldots 3$ (space). Signature $(-, +, +, +)$.
- Metrics: $g_{\mu\nu} = {}^{(4)}g_{\mu\nu}$, $\gamma_{ij} = {}^{(3)}g_{ij}$; flat spatial slices in Cartesian.
- 3+1: lapse α (set to 1), shift β^i, extrinsic curvature K_{ij}, with conventions as in [24–26].
- Covariant differentiation in space (in 3+1 decomposition) is denoted by D_i or D^i.
- The components of the Riemann curvature tensor ${}^{(4)}R^{\kappa}{}_{\mu\kappa\nu}$ and of the Ricci curvature tensor ${}^{(4)}R_{\mu\nu}$ are defined following the conventions in Misner et al. [23], where $R^{\alpha}{}_{\beta\mu\nu} = \partial_\mu \Gamma^{\alpha}_{\beta\nu} - \ldots$ and the Ricci tensor is ${}^{(4)}R_{\mu\nu} = {}^{(4)}R^{\kappa}{}_{\mu\kappa\nu}$.
- Tetrad components of tensors are denoted with carets (hats) above the indices: $K_{\hat{i}\hat{j}}$, as in [23].
- Tetrad components aligned with the eigenvectors of a tensor and whose magnitudes equal the corresponding eigenvalues are distinguished by enclosing the indices in parentheses: $T^{(\mu)}{}_{(\nu)}$.
- Units are denoted by square brackets, e.g., [m]. Geometrized (natural) units are employed in the analysis. SI units are used in a few physical examples. Specifically:
  - x, y, z, r, ρ: [m] in both geometrized and in SI.
  - $G, G^{00}, G_i{}^0, G^0{}_i$: [$\mathrm{m}^{-2}$] in both geometrized units and in SI units.
  - σ: [m^{-1}] in both geometrized and in SI.
  - $g_{\mu\nu}, g^{\mu\nu}, \gamma_{ij}, \gamma^{ij}, f, \theta, \phi$: [1] (dimensionless) in both geometrized and in SI.
  - v, c: [1] (dimensionless) in geometrized and [$\mathrm{m\,s}^{-1}$] in SI.
  - t: [m] in geometrized and [s] in SI.
  - κ: $= 8\pi G/c^4$: [1] (dimensionless) in geometrized; and [N^{-1}] in SI.
  - $T, T^{00} = \varrho, T_i{}^0, T^0{}_i$: [$\mathrm{m}^{-2}$] in geometrized and [$\mathrm{N\,m}^{-2}$] = [$\mathrm{J\,m}^{-3}$] in SI.
- We adopt Natário's [2] right-handed spherical coordinate system (r, θ, ϕ) (Fig. 1) with orthonormal basis vectors $\mathbf{e_r}$, $\mathbf{e}_\theta$, $\mathbf{e}_\phi$. The polar axis is the direction of travel: the x-axis. The coordinates are the radial distance $r = \sqrt{x^2 + y^2 + z^2} \in [0, \infty)$, the polar angle $\theta = \arccos(x/r) \in [0, \pi]$ measured from the positive x-axis, and the azimuthal angle $\phi = \mathrm{atan2}(z, y) \in [-\pi, \pi]$. Surfaces $\theta = \mathrm{const}$ define circular cones centered on the x-axis (vertex at origin), with $\theta = \pi/2$ being the yz (equatorial) plane. Constant r and θ define parallel circles centered on the x-axis, analogous to parallels of latitude on Earth. The cylindrical radius (distance from the x-axis) is $\mathsf{r} = r \sin\theta$. The spherical warp bubble boundary is defined by $r = \rho$. Its 'equator' (intersection with the yz-plane, $\theta = \pi/2$) is the largest of these parallels on the bubble surface and has cylindrical radius $\mathsf{r} = \rho \sin(\pi/2) = \rho$. The two-argument function $\mathrm{atan2}(z, y)$ computes the angle ϕ in the yz-plane measured from the positive y-axis to the projection of the position vector. Surfaces $\phi =$const are half-planes containing the x-axis; their intersection with the bubble boundary $r = \rho$ yields circles of radius ρ (analogous to meridians, i.e. lines of constant longitude). We refer to the xy-plane ($\phi = 0$) as the prime meridian plane.

Appendix B Regularity and asymptotics of the irrotational profile $g(r)$ at the origin

Axioms

(1) *Parameters:* $\sigma > 0$ and $\rho > 0$ are fixed. (2) *Analyticity:* Define

$$f_{\mathrm{Alc}}(r) := \frac{\tanh\big(\sigma(r+\rho)\big) - \tanh\big(\sigma(r-\rho)\big)}{2\,\tanh(\sigma\rho)}, \qquad f(r) := 1 - f_{\mathrm{Alc}}(r), \tag{B1}$$

and

$$g(r) \;:=\; \frac{1}{r}\int_0^r f(s)\,\mathrm{d}s \qquad (r \neq 0), \quad \text{with } g(0) \text{ defined by the limit } r \to 0. \tag{B2}$$

Since f_{Alc} is a composition/product of analytic functions, $f_{\mathrm{Alc}}, f \in C^\infty(\mathbb{R})$. Moreover,

$$f_{\mathrm{Alc}}(r) \;=\; \frac{\cosh^2(\sigma\rho)}{\cosh\big(\sigma(r+\rho)\big)\,\cosh\big(\sigma(r-\rho)\big)}, \tag{B3}$$

which is manifestly even in r.

Proposition

g is C^∞ near $r = 0$ and

$$\lim_{r\to 0} g(r) \;=\; 0, \qquad g(r) \;=\; \frac{\sigma^2\,\mathrm{sech}^2(\sigma\rho)}{3}\,r^2 \;+\; O(r^4) \quad (r \to 0). \tag{B4}$$

Proof. Step 1 (model–independent limit). By the Fundamental Theorem of Calculus, $\frac{\mathrm{d}}{\mathrm{d}r}\big(\int_0^r f(s)\,\mathrm{d}s\big) = f(r)$. Thus L'Hôpital's rule gives

$$\lim_{r\to 0} g(r) = \lim_{r\to 0} \frac{\int_0^r f(s)\,\mathrm{d}s}{r} = \lim_{r\to 0} f(r) = f(0). \tag{B5}$$

From (B1) and the oddness of tanh, $f_{\mathrm{Alc}}(0) = 1$ and hence

$$f(0) = 1 - f_{\mathrm{Alc}}(0) = 0. \tag{B6}$$

Therefore $\lim_{r\to 0} g(r) = 0$.

Step 2 (leading coefficient for f_{Alc}). Set $a := \sigma\rho$, $b := \sigma$ and write

$$F(r) := f_{\mathrm{Alc}}(r) = \frac{\cosh^2 a}{\cosh(a+br)\cosh(a-br)}.$$

Then

$$(\ln F)'(r) = -b\tanh(a+br)+b\tanh(a-br), \qquad (\ln F)''(r) = -b^2\operatorname{sech}^2(a+br)-b^2\operatorname{sech}^2(a-br),$$

so $(\ln F)'(0) = 0$ and $(\ln F)''(0) = -2b^2\operatorname{sech}^2 a$. Since $F'(0) = 0$, we have

$$f''_{\mathrm{Alc}}(0) = F''(0) = F(0)\,(\ln F)''(0) = -\,2\sigma^2\operatorname{sech}^2(\sigma\rho). \tag{B7}$$

Thus

$$f_{\mathrm{Alc}}(r) = 1 + \frac{f''_{\mathrm{Alc}}(0)}{2}\,r^2 + O(r^4) = 1 - \sigma^2\operatorname{sech}^2(\sigma\rho)\,r^2 + O(r^4), \tag{B8}$$

and hence

$$f(r) = 1 - f_{\mathrm{Alc}}(r) = \sigma^2\operatorname{sech}^2(\sigma\rho)\,r^2 + O(r^4). \tag{B9}$$

Integrating and dividing by r,

$$\begin{aligned} \int_0^r f(s)\,\mathrm{d}s &= \frac{\sigma^2\operatorname{sech}^2(\sigma\rho)}{3}\,r^3 + O(r^5), \\ g(r) &= \frac{1}{r}\int_0^r f(s)\,\mathrm{d}s = \frac{\sigma^2\operatorname{sech}^2(\sigma\rho)}{3}\,r^2 + O(r^4). \end{aligned} \tag{B10}$$

which proves (B4). □

Corollary (General limit for any C^1 form function).

If $f \in C^1$ near 0 and $f(0) = 0$, then $\lim_{r\to 0}\frac{1}{r}\int_0^r f(s)\,\mathrm{d}s = 0$. *Proof.* Either by the Fundamental Theorem of Calculus and L'Hôpital's rule (as in Step 1) or by the integral mean value theorem: $\frac{1}{r}\int_0^r f = f(\xi) \to f(0) = 0$ for some $\xi \in (0, r)$. □

Remark 1 (Evenness and absence of a linear term)

From (B3), f_{Alc} is even; thus $f = 1-f_{\mathrm{Alc}}$ is even, $f'(0) = 0$, and $f(r) = O(r^2)$, yielding $g(r) = O(r^2)$ by (B10). This reflects axial symmetry about the direction of motion.

Remark 2 (Integration constant)

Definition (B2) fixes g. Any primitive $\tilde{g}(r) = \frac{1}{r}\big(\int_0^r f + C\big)$ is regular at $r = 0$ only if $C = 0$; otherwise $\tilde{g}(r) \sim C/r$.

Remark 3 (Hypothetical linear term and radial C^1 regularity)

If one posits $f(r) = a\,r + O(r^2)$ for small $r > 0$, then $\frac{1}{r}\int_0^r f(s)\,\mathrm{d}s = \frac{a}{2}r + O(r^2) \to 0$, so $g(0) = 0$ still holds. However, such a linear term is *incompatible with a C^1, rotationally symmetric scalar field on* $\mathbb{R}^3$: writing $u(\mathbf{x}) = \phi(|\mathbf{x}|)$, differentiability at $\mathbf{0}$ forces $\phi'(0) = 0$ (otherwise directional derivatives at $\mathbf{0}$ would be a nonzero constant, contradicting linearity of $v \mapsto \nabla u(0)\cdot v$ on the unit sphere). In our model, $f = 1 - f_{\mathrm{Alc}}$ is even by construction (cf. (B3)).

Appendix C Global Energy Budget, Volumes, Tail Model, and Localization Diagnostics

This appendix quantifies the global negative– and positive–energy budgets and their spatial distribution on the numerical window used for integration, and then attaches a physics– and numerics–based tail correction to estimate the contribution from $r > R_{\text{integ}}$.

Field, masks, and nondimensionalization

We compute the spatial Hessian $H_{ij} = \partial_i \partial_j \Phi$ and the rotationally invariant scalar

$$\lambda_H \;=\; \tfrac{1}{2}\Big(\mathrm{Tr}[H^2] - \big(\mathrm{Tr}\, H\big)^2\Big)\,, \tag{C.1}$$

which, for an irrotational shift $\beta_i = \partial_i \Phi$ on a stationary flat slice, obeys

$$K_{ij}K^{ij} - K^2 \;=\; 2\,\lambda_H \;= -2\kappa\,\varrho_p, \tag{C.2}$$

so that the proper energy density follows from Einstein's equations as

$$\varrho_p \;=\; -\,\frac{\lambda_H}{\kappa}, \qquad \kappa = 8\pi \text{ (geom.)}, \quad \kappa = 2.07665 \times 10^{-43}\ [\mathrm{N}^{-1}] \text{ (SI)}. \tag{C.3}$$

Basis and notation.

Indices i, j are *Cartesian* on the spatial slice. In an orthonormal tetrad, $T^{(0)}{}_{(0)} = -\varrho_p$. Using $G^{(a)}{}_{(b)} = \kappa\, T^{(a)}{}_{(b)}$ one has $\lambda_{G(0)} = \kappa\,\lambda_{T(0)} = -\kappa\,\varrho_p = \lambda_H$, hence $\varrho_p = -\lambda_H/\kappa$.

Independent computation of the timelike eigenvalue.

Besides extracting the timelike eigenvalue $\lambda_{G(0)}$ in the main text by direct diagonalization of the mixed Einstein tensor,

$$G^{(a)}{}_{(b)}\, u^{(b)}_{(0)} = \lambda_{G(0)}\, u^{(a)}_{(0)}, \tag{C.4}$$

we compute it here independently, from the Hessian route developed above. Using Einstein's equations $G^{(a)}{}_{(b)} = \kappa\, T^{(a)}{}_{(b)}$ and the invariant $\lambda_H = \frac{1}{2}\big(\mathrm{Tr}[H^2] - (\mathrm{Tr}\, H)^2\big)$, the timelike eigenvalues of $G^{(a)}{}_{(b)}$ and $T^{(a)}{}_{(b)}$ satisfy

$$\lambda_{G(0)} = \kappa\,\lambda_{T(0)} = -\kappa\,\varrho_p = \lambda_H. \tag{C.5}$$

Thus the timelike eigenvalue obtained by direct diagonalization (Eq. (C.4)) is identically reproduced by the Hessian construction λ_H without any eigenvector ordering or degeneracy issues. Numerically, over the reported grids and working precision, the two evaluations of $\lambda_{G(0)}$ agree to roundoff, providing a robust internal consistency check on `Eigensystem` and on the analytic–derivative implementation used to build H_{ij}.

Disjoint masks and tolerances.

We partition space by the sign of λ_H with absolute deadbands:

$$\begin{aligned} \mathscr{R}_- &= \{\lambda_H > \lambda_{\rm tol}^{(+)}\}, \qquad \mathscr{R}_+ = \{\lambda_H < -\lambda_{\rm tol}^{(-)}\}, \\ \mathscr{R}_0 &= \{\, |\lambda_H| \le \max\!\big(\lambda_{\rm tol}^{(+)}, \lambda_{\rm tol}^{(-)}\big) \,\}, \end{aligned} \tag{C.6}$$

$$\lambda_{\rm tol}^{(\pm)} \;=\; \max\!\big\{\lambda_{\rm floor},\; \varepsilon\, \lambda_{\rm max}^{(\pm)}\big\}, \qquad \lambda_{\rm floor} = 10^{-20}, \;\; \varepsilon = 10^{-6}. \tag{C.7}$$

Grid, volume element, and numerical window

With axial symmetry the azimuthal angle is integrated analytically, so

$$dV \;=\; 2\pi\, r^2 \sin\theta \; dr \; d\theta. \tag{C.8}$$

We use a uniform tensor grid $(n_r, n_\theta) = (1601, 1441)$ at 40-digit working precision. The integration window is $r \in [0, R_{\rm integ}]$ with $R_{\rm integ} = 12\,\rho$. A posteriori checks (Eq. (C.18)) confirm the Jacobian normalization and mask disjointness to machine accuracy.

Reporting precision.

Consistent with common practice, we round *central values* to ∼3 significant figures (s.f.) unless otherwise noted; *percentages* are reported to two decimals; and *uncertainties* are quoted with 1–2 s.f., with central values rounded to the same decimal place as their uncertainties.

Nondimensionalization.

For the irrotational profile, analytic structure gives $\lambda_H \sim v^2\sigma^2\, F(\sigma r, \theta)$. We therefore present the dimensionless field (for $\sigma v \neq 0$)

$$\widetilde{\varrho}_p \;\equiv\; \frac{\kappa\, \varrho_p}{\sigma^2 (v/c)^2} \;=\; -\,\frac{\lambda_H}{\sigma^2 (v/c)^2}, \tag{C.9}$$

on dimensionless axes (for $\rho \neq 0$)

$$(\tilde{x}, \tilde{y}) \;=\; \Big(\frac{x}{\rho}, \frac{y}{\rho}\Big), \tag{C.10}$$

so that runs with the same $\sigma\rho$ therefore collapse onto a single nondimensional map, while the warp-bubble wall remains near $r/\rho \simeq 1$. .

Prime meridional plane energy density map

On the plotted (zoomed) disk of Fig. C1 with $R_{\rm plot} = 2\rho$, the SI extrema are $\min \varrho_p = -1.29 \times 10^{41}\,[\mathrm{J\,m^{-3}}]$ and $\max \varrho_p = 2.23 \times 10^{42}\,[\mathrm{J\,m^{-3}}]$. These extrema are governed by the near–wall structure. By construction, blue indicates $\varrho_p < 0$ and red indicates $\varrho_p > 0$; the solid black curve marks the zero locus $\varrho_p = 0$ (equivalently $\lambda_H = 0$).

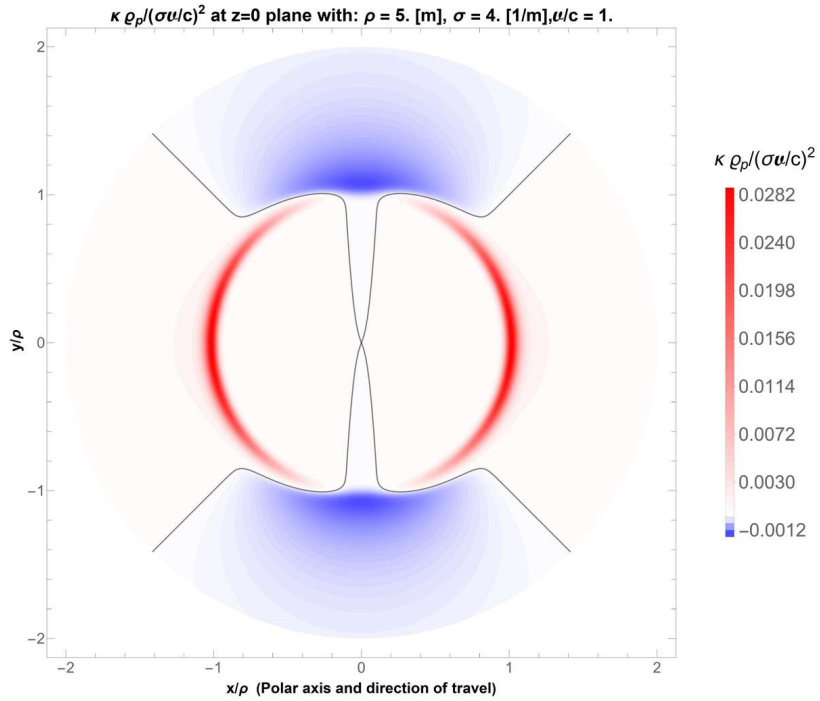


Fig. C1: Prime meridional slice ($z = 0$, xy-plane) of the nondimensionalized proper energy density $\kappa\, \varrho_p/(\sigma v/c)^2$ versus nondimensional axes x/ρ (polar axis and direction of motion) and y/ρ (transverse in–plane axis). The solid black curve is the null energy density $\varrho_p = 0$ contour separating $\varrho_p < 0$ (blue) from $\varrho_p > 0$ (red). Parameters: $\rho = 5$ [m], $\sigma = 4$ [m^{-1}], $v/c = 1$.

For the representative run $\rho = 5$ [m], $\sigma = 4$ [m^{-1}], $v/c = 1$ (so $\sigma\rho = 20$), the *positive* energy density on the prime meridional plane appears as two *disjoint, crescent–shaped* lobes concentrated near $r \simeq \rho$. The crescents are centered along the $\pm x$ directions (axis of motion), where ϱ_p attains its in–plane maxima, and they taper toward the $\pm y$ directions at the equator, where the $\varrho_p = 0$ contour pinches the would–be annulus into two separate lobes. The configuration exhibits twofold (C_2) rotational symmetry about the bubble center, i.e. invariance under the half–turn $(x, y) \mapsto (-x, -y)$. Each crescent has characteristic half–thickness $\sim 1/\sigma$, which on the nondimensional axes corresponds to $\Delta(x/\rho) \sim \Delta(y/\rho) \sim 1/(\sigma\rho) \approx 0.05$.

The *negative* energy density appears (i) outside the bubble ($r > \rho$) as weak exterior caps that occupy approximately quarter–disk sectors, whose $\varrho_p = 0$ boundaries are straight rays at $\pm 45°$ to the x axis; and (ii) inside the bubble ($r < \rho$) as thin, very low amplitude slivers aligned with the y axis.

3D interpretation. Revolving the two positive–energy crescents about the x–axis (polar axis) produces two *disjoint polar collars* on the thin spherical shell $r \simeq \rho$, with maximal thickness at the poles ($\theta = 0, \pi$) and vanishing near the equator ($\theta = \pi/2$). In contrast, the exterior negative–energy wedges lie at slightly larger radius ($r > \rho$)

and, upon revolution, sweep out broad *equatorial mantles* (annular sectors): they are low in amplitude but extend substantially in radius, peaking near the equator and decaying with increasing r.

Why the volumes differ.

Although the $z = 0$ meridional slice shows comparable positive and negative ϱ_p areas—which approach equality as $r \to \infty$ because the $\varrho_p = 0$ contours are straight rays at $\pm 45°$ to the x-axis for $r > \rho$—the axisymmetric 3-D volume element weights each (r, θ)-ring by $2\pi r^2 \sin\theta$ (Eq. (C.8)). The positive support is confined to thin polar collars near $r \simeq \rho$ at small $\sin\theta$, whereas the negative support occupies broad equatorial mantles extending to larger r where $\sin\theta \approx 1$; hence $V_- > V_+$ even when the $z = 0$ slice areas appear similar (cf. Fig. C1, Tables C4–C5).

Global energy magnitudes and volumes (baseline window $R_{\text{integ}} = 12\rho$)

Let λ_H be as in Eq. (C.1). By $\kappa\, \varrho_p = -\lambda_H$,

$$E_- = \int_{\mathscr{R}_-} |\varrho_p|\, dV = \frac{1}{\kappa} \int_{\mathscr{R}_-} |\lambda_H|\, dV, \qquad E_+ = \int_{\mathscr{R}_+} |\varrho_p|\, dV = \frac{1}{\kappa} \int_{\mathscr{R}_+} |\lambda_H|\, dV,$$
$$E_{\text{neg}} = -E_-, \quad E_{\text{pos}} = E_+, \quad E_{\text{net}} = E_+ - E_-. \tag{C.11}$$

Region volumes are

$$V_- = \int_{\mathscr{R}_-} dV, \qquad V_0 = \int_{\mathscr{R}_0} dV, \qquad V_+ = \int_{\mathscr{R}_+} dV, \qquad V_{\text{act}} = V_- + V_0 + V_+. \tag{C.12}$$

Baseline numerical totals ($R_{\text{integ}} = 12\rho$, no tail).

See Tables C2 and C3.

Large–radius tail model and two–point estimator

Asymptotic scaling and why $1/R$.

At large r, the analytic form used in the code gives $g(r) = 1 - \dfrac{I}{r} + \mathscr{O}\big(e^{-2\sigma r}/r\big)$ with $I = \rho \coth(\sigma\rho)$. Then $\Phi = v\, r\, g(r) \cos\theta = v\,(x - I\cos\theta) + \mathscr{O}(e^{-2\sigma r})$, so the Hessian obeys $H_{ij} = \partial_i \partial_j \Phi = \mathscr{O}(r^{-2})$ and the invariant $\lambda_H = \frac{1}{2}\big(\mathrm{Tr}[H^2] - (\mathrm{Tr}\, H)^2\big) = \dfrac{C(\theta)}{r^4} + \mathscr{O}(e^{-4\sigma r})$, with $C(\theta) \geq 0$ on a set of nonzero solid angle. With an *absolute* deadband (Eq. (C.7)) and over the radii used for fitting ($8\rho \leq r \leq 12\rho$), the active solid–angle fraction where $|\lambda_H| > \lambda_{\text{tol}}^{(+)}$ is *order–unity* (cf. Table C4). Hence a shell contributes $dE_-(r) \propto (r^{-4}) \times r^2\, dr = r^{-2} dr$, and the far–field tail satisfies $E_-(\infty) - E_-(R) \sim \displaystyle\int_R^\infty r^{-2} dr \sim 1/R$. (For E_+, support is compact near $r \simeq \rho$, so the tail is much smaller—matching the data.)

Two–point $1/R$ estimator.

If $E_\pm(\infty) - E_\pm(R) = A_\pm/R + \mathscr{O}(R^{-2})$, then for $R_1 < R_2$

$$E_\pm(\infty) \;\approx\; \frac{R_2\,E_\pm(R_2) - R_1\,E_\pm(R_1)}{R_2 - R_1} \;=\; E_\pm(R_2) \;+\; \frac{R_1}{R_2 - R_1}\Big(E_\pm(R_2) - E_\pm(R_1)\Big). \tag{C.13}$$

We use $R_1 = 8\rho$, $R_2 = 12\rho$, so the tail increment is $\Delta E_\pm^{\text{tail}} = E_\pm(\infty) - E_\pm(12\rho) = 2\,[E_\pm(12\rho) - E_\pm(8\rho)]$.

Tail–corrected totals (two–point).

Using Eq. (C.13) with $R_1 = 8\rho$, $R_2 = 12\rho$ and the measured $\{E_\pm(8\rho), E_\pm(12\rho)\}$, we obtain

$$\Delta E_-^{\text{tail}} = 1.19 \times 10^{43}\ [\text{J}], \qquad \Delta E_+^{\text{tail}} = 3.42 \times 10^{42}\ [\text{J}],$$

and

$$E_-(\infty) = 1.33 \times 10^{44}\ [\text{J}], \quad E_+(\infty) = 1.33 \times 10^{44}\ [\text{J}], \quad E_{\text{net}}(\infty) = -1.12 \times 10^{41}\ [\text{J}]. \tag{C.14}$$

Thus $E_{\text{abs}}(\infty) = E_-(\infty) + E_+(\infty) = 2.66 \times 10^{44}$ [J] and $|E_{\text{net}}|/E_{\text{abs}}(\infty) = 0.04\%$.

Table C1: Baseline ($R_{\text{integ}} = 12\rho$) versus tail–corrected ($R \to \infty$) global energies using a two–point $1/R$ extrapolation with $R_1 = 8\rho$, $R_2 = 12\rho$.

Quantity	Baseline at 12ρ	Tail–corrected $R \to \infty$
E_- [J]	1.21×10^{44}	1.33×10^{44}
E_+ [J]	1.29×10^{44}	1.33×10^{44}
$E_{\text{net}} = E_+ - E_-$ [J]	8.37×10^{42}	-1.12×10^{41}
$\dfrac{\lvert E_{\text{net}}\rvert}{\int \lvert\varrho_p\rvert\, dV}$ [%]	3.34	0.04

Numerically, the tail raises both magnitudes and drives the net toward zero: $|E_{\text{net}}|/\int |\varrho_p|\, dV$ drops from 3.34% (baseline window) to 0.04% (tail–corrected).

The near–cancellation $E_+ \approx E_-$ thus becomes extremely tight once the far–field is included.

Physical remark on near cancellation of net energy.

The *proper* energy magnitudes nearly cancel: on the tail–corrected totals we find $|E_{\text{net}}|/E_{\text{abs}} = 0.04\%$ (two–point $1/R$ extrapolation; percentages reported to two decimals). This statement concerns the *proper* energy density defined by the timelike eigenvalue of $T^\mu{}_\nu$; it does not, by itself, establish a vanishing ADM or Komar mass, which require global geometric fluxes/constraints beyond the local energy density.

Localization diagnostics (baseline window)

Equatorial band captures about $\theta = \pi/2$***.***

For half–width Δ,

$$
\begin{aligned}
f_{V_\pm}^{\text{band}}(\Delta) &= \frac{\displaystyle\int_{\mathscr{R}_\pm}\int_{\pi/2-\Delta}^{\pi/2+\Delta} 2\pi r^2 \sin\theta \, dr \, d\theta}{V_\pm}, \\
f_{E_\pm}^{\text{band}}(\Delta) &= \frac{\displaystyle\int_{\mathscr{R}_\pm}\int_{\pi/2-\Delta}^{\pi/2+\Delta} \frac{|\lambda_H|}{\kappa} 2\pi r^2 \sin\theta \, dr \, d\theta}{E_\pm}.
\end{aligned}
\tag{C.15}
$$

Bubble wall–shell captures.

For half–width w/σ about $r = \rho$ (clipped to $[0, R_{\text{integ}}]$),

$$
\begin{aligned}
f_{V_\pm}^{\text{shell}}(w) &= \frac{\displaystyle\int_{\mathscr{R}_\pm}\int_{\rho-w/\sigma}^{\rho+w/\sigma} 2\pi r^2 \sin\theta \, dr \, d\theta}{V_\pm}, \\
f_{E_\pm}^{\text{shell}}(w) &= \frac{\displaystyle\int_{\mathscr{R}_\pm}\int_{\rho-w/\sigma}^{\rho+w/\sigma} \frac{|\lambda_H|}{\kappa} 2\pi r^2 \sin\theta \, dr \, d\theta}{E_\pm}.
\end{aligned}
\tag{C.16}
$$

Numerical validation

Unmasked signed integral reproduces the masked net energy to within roundoff:

$$
E_{\text{net}}^{(\text{unmasked})} = \int \left(-\frac{\lambda_H}{\kappa} \right) dV \;\approx\; E_{\text{net}}^{(\text{masked})}. \tag{C.17}
$$

Masks partition the grid and the discrete volume form is consistent:

$$
\max_{\text{grid}} \left| \mathbf{1}_{\mathscr{R}_-} + \mathbf{1}_{\mathscr{R}_0} + \mathbf{1}_{\mathscr{R}_+} - 1 \right| = 0, \qquad \sum_{\text{grid}} dV / V_{\text{plot}} = 1.000937, \tag{C.18}
$$

indicating a relative volume-normalization error of 9.37×10^{-4}.

Numerical summary for $\boldsymbol{\rho = 5}$ [m], $\boldsymbol{\sigma = 4}$ [$\mathbf{m^{-1}}$], $\boldsymbol{v/c = 1}$

Tail–corrected net. Using the two–point $1/R$ extrapolation, **the net proper energy satisfies** $|E_{\text{net}}(\infty)|/E_{\text{abs}}(\infty) = 0.04\%$, i.e. **effectively zero at the reported precision.**

Table C2: Global energy magnitudes and volumes (baseline window $R_{\text{integ}} = 12\rho$).

Quantity	Value
Negative energy E_- [J]	1.21×10^{44}
Positive energy E_+ [J]	1.29×10^{44}
Net energy $E_{\text{net}} = E_+ - E_-$ [J]	8.37×10^{42}
$\int \lvert\varrho_p\rvert \, dV$ [J]	2.50×10^{44}
Negative volume V_- [m^3]	6.38×10^5
Positive volume V_+ [m^3]	2.34×10^5
Zero volume V_0 [m^3]	3.40×10^4

Table C3: Ratios and consistency checks (baseline window).

Metric	Value
Energy ratio E_+/E_-	1.07
$\lvert E_{\text{net}}\rvert / \int \lvert\varrho_p\rvert \, dV$	3.34%
$V_-/(V_- + V_+ + V_0)$	70.44%
$V_+/(V_- + V_+ + V_0)$	25.81%
$V_0/(V_- + V_+ + V_0)$	3.75%
$V_-/(V_- + V_+)$	73.19%
$V_+/(V_- + V_+)$	26.81%
V_-/V_+	272.95%
Direct signed sum $E_{\text{net}}^{\text{direct}}$ [J]	8.40×10^{42}
$\lvert E_{\text{net}} - E_{\text{net}}^{\text{direct}}\rvert / \int \lvert\varrho_p\rvert \, dV$	0.0138%

Table C4: Prime meridional band captures about $\theta = \pi/2$ (baseline window).

Δ	V_-	E_-	V_+	E_+
5°	12.52%	19.53%	0.00%	0.00%
10°	24.79%	37.77%	0.00%	0.02%
15°	36.87%	54.24%	0.01%	0.28%
20°	48.67%	68.50%	0.02%	1.18%

Appendix D Causal Proof: A Fixed-Smoothing Vortical Ablation Study

Goal. To isolate the causal role of *irrotationality*, we conduct a rigorous ablation study. We hold the baseline "smoothing" profile—derived from the scalar potential Φ and its associated integral map $g(r)$—*fixed*, and then introduce a tunable vortical component. By recomputing the energetics with the *same* numerical grid, masks, and tolerances, any observed degradation in the energy metrics is attributable solely to the loss of the curl-free condition, not to a change in smoothing.

Table C5: Bubble wall–shell captures for $|r-\rho| \le w/\sigma$ (baseline window).

w	V_-	E_-	V_+	E_+
0.5	0.00%	0.33%	0.03%	34.24%
1.0	0.00%	1.09%	0.06%	58.43%
2.0	0.01%	3.43%	0.11%	71.21%
3.0	0.02%	7.52%	0.14%	74.42%

Shared assumptions and numerics

We work on a stationary, flat spatial slice with unit lapse $\alpha = 1$. All physical parameters ($\rho = 5$ [m], $\sigma = 4$ [m^{-1}], $v/c = 1$), the integration window ($r \in [0, 12\rho]$), and the numerical grid $(n_r, n_\theta) = (1601, 1441)$ are identical to the baseline irrotational case. Crucially, the masks $\mathscr{R}_\pm, \mathscr{R}_0$ and their absolute tolerances $\lambda_{\rm tol}^{(\pm)}$ are *frozen* to the baseline values (see Appendix C, Eqs. (C.6)–(C.7)), ensuring a consistent and fair comparison across the entire ablation sweep. Central values are reported to 3 significant figures and percentages to two decimals, matching the conventions used elsewhere.

Kinematics and tensors

Baseline irrotational structure

The baseline metric is purely irrotational, with $\beta_i = \partial_i \Phi$ and $\Phi = v\, r\, g(r) \cos\theta$. The extrinsic curvature is the Hessian of the potential,

$$K_{ij} \;=\; H_{ij} := \partial_i \partial_j \Phi, \tag{D.1}$$

As established in Appendix C, the Hessian invariant λ_H and the proper energy density ϱ_p are linked through Eqs. (C.1) and (C.3).

Vortical ablation with fixed smoothing

We add a vortical component to the baseline shift vector,

$$\beta \;=\; \nabla\Phi \;+\; \nabla \times \mathscr{A}, \tag{D.2}$$

where the vorticity is introduced via a toroidal vector potential $\mathscr{A} = \mathscr{A}_\phi(r,\theta)\, \mathbf{e}_\phi$ with

$$\mathscr{A}_\phi(r,\theta) \;=\; \psi(r)\, \sin\theta. \tag{D.3}$$

The radial profile $\psi(r)$ generates a vortical perturbation localized in a compact shell around the bubble wall:

$$\psi(r) \;=\; \eta\, v \left(\tfrac{r}{\rho}\right)^2 \exp\!\left[-\left(\tfrac{r-\rho}{w/\sigma}\right)^2\right], \qquad w = 1, \quad \eta \in \{0,\, 0.25,\, 0.5,\, 0.75,\, 1\}. \tag{D.4}$$

In an orthonormal spherical tetrad $(\hat{r}, \hat{\theta}, \hat{\phi})$, the nonzero components of the full extrinsic curvature K are

$$K_{\hat{r}\hat{r}}(r,\theta) = \left[v\,(2g'(r) + r\,g''(r)) + 2\left(\tfrac{\psi'(r)}{r} - \tfrac{\psi(r)}{r^2}\right)\right] \cos\theta, \tag{D.5}$$

$$K_{\hat{\theta}\hat{\theta}}(r,\theta) = \left[v\,g'(r) - \tfrac{\psi'(r)}{r} + \tfrac{\psi(r)}{r^2}\right] \cos\theta, \qquad K_{\hat{\phi}\hat{\phi}}(r,\theta) = K_{\hat{\theta}\hat{\theta}}(r,\theta), \tag{D.6}$$

$$K_{\hat{r}\hat{\theta}}(r,\theta) = -\left[v\,g'(r) + \tfrac{1}{2}\,\psi''(r)\right] \sin\theta, \qquad K_{\hat{r}\hat{\phi}}(r,\theta) = K_{\hat{\theta}\hat{\phi}}(r,\theta) = 0. \tag{D.7}$$

The quadratic invariant for the ablation, which generalizes Eqs. (C.1) and (C.3) (and reduces to those relations when $\eta = 0$), is

$$\lambda_{\text{blend}}(r,\theta) = \tfrac{1}{2}\left(K_{ij}K^{ij} - K^2\right), \qquad K := K^i{}_i = K_{\hat{r}\hat{r}} + K_{\hat{\theta}\hat{\theta}} + K_{\hat{\phi}\hat{\phi}}, \tag{D.8}$$

and in terms of the orthonormal components it reads

$$\lambda_{\text{blend}} = \tfrac{1}{2}\left(K_{\hat{r}\hat{r}}^2 + 2\,K_{\hat{\theta}\hat{\theta}}^2 + 2\,K_{\hat{r}\hat{\theta}}^2 - \left(K_{\hat{r}\hat{r}} + 2\,K_{\hat{\theta}\hat{\theta}}\right)^2\right), \qquad \kappa\,\varrho_p = -\,\lambda_{\text{blend}}. \tag{D.9}$$

Baseline equality at $\eta = 0$

To ensure bit-for-bit consistency, the $\eta = 0$ case reuses the stored baseline grid $\lambda_H(r,\theta)$, guaranteeing that all baseline totals are reproduced to roundoff:

$$\eta = 0 \quad \Longrightarrow \quad \lambda_{\text{blend}}(r,\theta) \equiv \lambda_H(r,\theta). \tag{D.10}$$

Global measures

Global magnitudes and volumes follow Appendix C, Eqs. (C.11)–(C.12), with λ replaced by λ_{blend} in the ablation.

Numerical results (3 s.f.)

Windowed sweep at $R_{\text{integ}} = 12\rho$

Table D6 reports the global magnitudes $E_- = \int_{\mathscr{R}_-} |\varrho_p|\,dV$, $E_+ = \int_{\mathscr{R}_+} |\varrho_p|\,dV$, and the ratio E_+/E_- for $\eta \in \{0, 0.25, 0.5, 0.75, 1\}$. The mask tolerances are held fixed to their baseline values.

Interpretation

Because $g(r)$ is held fixed and the mask tolerances are frozen to the baseline values, Table D6 shows that even a modest vortical admixture ($\eta = 0.25$) increases the negative-energy magnitude E_- by a factor of ≈ 32 relative to baseline, while collapsing the balance E_+/E_- by nearly two orders of magnitude (from 1.07 to 1.53×10^{-2}). As η increases to 1, E_- grows by a factor $> 5 \times 10^2$ and E_+/E_- falls to 5.84×10^{-4}. **This demonstrates that the favorable energy properties reported in the main**

Table D6: Fixed-smoothing vortical ablation sweep on the baseline window $R_{\text{integ}} = 12\rho$. Central values to 3 s.f.

η	E_- [J]	E_+ [J]	E_+/E_-
0.00	1.21×10^{44}	1.29×10^{44}	1.07
0.25	3.92×10^{45}	6.00×10^{43}	1.53×10^{-2}
0.50	1.55×10^{46}	4.91×10^{43}	3.16×10^{-3}
0.75	3.49×10^{46}	4.13×10^{43}	1.18×10^{-3}
1.00	6.19×10^{46}	3.62×10^{43}	5.84×10^{-4}

text—particularly the near-cancellation of integrated proper energy—are a direct consequence of the irrotational, curl-free kinematics and not an artifact of profile smoothing.

Appendix E Boundary behaviour details

The limiting values of the shift components are summarized in Table E7. For Alcubierre, the interior shift equals $+v(t)$ along $+x$ and decays for large r. For Natário and for the irrotational model (Sec. 3.4.2), the bubble center is at rest while the external universe drifts with velocity $-v(t)$, consistent with the adopted convention.

Model	Angle	$\lim_{r\to 0} \beta^{\hat{r}}$	$\lim_{r\to 0} \beta^{\hat{\theta}}$	$\lim_{r\to \infty} \beta^{\hat{r}}$	$\lim_{r\to \infty} \beta^{\hat{\theta}}$
Alcubierre	$\theta = 0$	$v(t)$	0	0	0
	$\theta = \frac{\pi}{2}$	0	$-v(t)$	0	0
Natário	$\theta = 0$	0	0	$-v(t)$	0
	$\theta = \frac{\pi}{2}$	0	0	0	$v(t)$
Irrotational	$\theta = 0$	0	0	$-v(t)$	0
	$\theta = \frac{\pi}{2}$	0	0	0	$v(t)$

Table E7: Boundary behaviour of the shift tetrad components for the Alcubierre, Natário, and irrotational drives: limiting values of $(\beta^{\hat{r}}, \beta^{\hat{\theta}})$ at $r \to 0$ and $r \to \infty$ for two representative polar angles. (In all three, $\beta^{\hat{\phi}} = 0$ by axial symmetry.)

Appendix F Canonical Type-IV blocks and parameter maps—Relation to the literature

Table F8 aligns the 2×2 blocks used in the literature with the mixed canonical form $T^a{}_b = \left(\begin{smallmatrix} \xi & \mu \\ -\mu & \xi \end{smallmatrix}\right)$. Apparent differences come only from index position (covariant/contravariant), basis ordering, and signature; after the appropriate raising/lowering and

Table F8: Canonical Type–IV 2×2 blocks across the literature and mapping to our mixed canonical form $T^a{}_b = \begin{pmatrix} \xi & \mu \\ -\mu & \xi \end{pmatrix}$.

Source	Initial tensor block	Basis / signature	Mixed block	Eigenvalues	Mapping to (ξ, μ)
This work	$T^a{}_b = \begin{pmatrix} \xi & \mu \\ -\mu & \xi \end{pmatrix}$	Ort. (t,s) / $(-,+,+,+)$	$T^a{}_b = \begin{pmatrix} \xi & \mu \\ -\mu & \xi \end{pmatrix}$	$\xi \pm i\mu$	$\xi = \xi,\ \mu = \mu$
Hawking–Ellis	$T^{ab} = \begin{pmatrix} -\varkappa & \nu \\ \nu & 0 \end{pmatrix}$	Ort. (s,t) / $(+,+,+,-)$	$T^a{}_b = \begin{pmatrix} -\varkappa & -\nu \\ \nu & 0 \end{pmatrix}$	$-\varkappa/2$ $\pm\frac{i}{2}\sqrt{4\nu^2 - \varkappa^2}$	$\xi = -\varkappa/2$ $\mu = \sqrt{\nu^2 - \varkappa^2/4}$
Maeda–Martinez	$T^{ab} = \begin{pmatrix} \rho & \nu \\ \nu & -\rho \end{pmatrix}$	Ort. (t,s) / $(-,+,+,+)$	$T^a{}_b = \begin{pmatrix} -\rho & \nu \\ -\nu & -\rho \end{pmatrix}$	$-\rho \pm i\nu$	$\xi = -\rho,\ \mu = \nu$
Moruno–Visser	$T^{ab} = \begin{pmatrix} \rho & f \\ f & -\rho \end{pmatrix}$	Ort. (t,s) / $(-,+,+,+)$	$T^a{}_b = \begin{pmatrix} -\rho & f \\ -f & -\rho \end{pmatrix}$	$-\rho \pm if$	$\xi = -\rho,\ \mu = f$
Petrov	$T_{ab} = \begin{pmatrix} \alpha-\beta & \alpha+\beta \\ \alpha+\beta & -\alpha+\beta \end{pmatrix}$	Qs. (k,l) / $g_{ab} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$	$T^a{}_b = \begin{pmatrix} \alpha & \beta \\ -\beta & \alpha \end{pmatrix}$	$\alpha \pm i\beta$	$\xi = \alpha,\ \mu = \beta$

basis change, all reduce to the same mixed block with conjugate eigenvalues $\xi \pm i\mu$ and an invariant null two–plane $g(z, \bar{z}) = 0$.

Mappings (see Table F8): We use the mixed canonical form $T^a{}_b = \begin{pmatrix} \xi & \mu \\ -\mu & \xi \end{pmatrix}$ in an orthonormal (t, s) basis with signature $(-,+,+,+)$. Hawking–Ellis [19, p. 90] start from $T^{ab} = \begin{pmatrix} -\varkappa & \nu \\ \nu & 0 \end{pmatrix}$ (orthonormal (s,t); $(+,+,+,-)$). Lowering yields $T^a{}_b = \begin{pmatrix} -\varkappa & -\nu \\ \nu & 0 \end{pmatrix}$ with eigenvalues $\frac{1}{2}\left(-\varkappa \pm i\sqrt{4\nu^2 - \varkappa^2}\right)$, hence $\xi = -\varkappa/2$, $\mu = \sqrt{\nu^2 - \varkappa^2/4}$; the Type IV condition is $\varkappa^2 < 4\nu^2$. Maeda–Martínez [21] use $T^{ab} = \begin{pmatrix} \rho & \nu \\ \nu & -\rho \end{pmatrix}$ ((t,s); $(-,+,+,+)$), giving $T^a{}_b = \begin{pmatrix} -\rho & \nu \\ -\nu & -\rho \end{pmatrix}$ with $-\rho \pm i\nu$, so $\xi = -\rho$, $\mu = \nu$. Martín–Moruno–Visser [33] have $T^{ab} = \begin{pmatrix} \rho & f \\ f & -\rho \end{pmatrix}$ (same signature), hence $T^a{}_b = \begin{pmatrix} -\rho & f \\ -f & -\rho \end{pmatrix}$ with $-\rho \pm if$, so $\xi = -\rho$, $\mu = f$. Petrov [40, §47] writes the covariant block $T_{ab} = \begin{pmatrix} \alpha-\beta & \alpha+\beta \\ \alpha+\beta & -\alpha+\beta \end{pmatrix}$ in a quasi–orthonormal (k, l) frame with $g_{ab} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$; raising gives $T^a{}_b = \begin{pmatrix} \alpha & \beta \\ -\beta & \alpha \end{pmatrix}$ with $\alpha \pm i\beta$, so $\xi = \alpha$, $\mu = \beta$.

All four descriptions thus encode the same algebraic geometry. The invariant null two–plane does *not* imply a real null eigenvector, consistent with [19]. The "invariant-null-plane $\Rightarrow$ null-eigenvector" lemma of Stephani et al. [37, p. 58, item (4)] is proved for $R^a{}_b$; since $G^a{}_b = R^a{}_b - \frac{1}{2}R\,\delta^a{}_b$ differs only by a trace, eigenvectors and invariant planes are unchanged, and the conclusion transfers verbatim to $G^a{}_b \propto T^a{}_b$.

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